MBA Thesis

A Comparative Cost Model for Sheet Metal Profile Manufacturing Strategy

Break-Even Analysis of Roll Forming and Profile Bending Across Investment Levels

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The authors declare that they have completed the thesis work independently. All external sources are cited and listed under the References section. The thesis work has not been submitted in the same or similar form to any other institution(s) as part of another examination or degree.

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Abstract

Today's manufacturing companies encounter complex strategic dilemmas when making investments in production processes, due to global market demands for both cost efficiency and product variety. This thesis develops and applies a quantitative, comparative cost model to evaluate operational performance and to determine the economic break-even points between so called roll forming and profile bending, two leading technologies within the segment of sheet metal profile fabrication.

Through utilising empirical research design, the model presented in this thesis integrates machine supplier data and primary operational figures in a production scenario featuring seven distinct sheet metal profile types constituting a product mix with volumes representative of yearly market demand. Both of the manufacturing technologies are analysed across three equipment investment configurations: Small, Medium, and Large. The cost modelling incorporates initial capital expenditures, variable production inputs such as material and energy, cycle-time with setup and changeover parameters, as well as inventory holding costs calculated using the Economic Order Quantity (EOQ) framework.

The resulting cost curves indicate that roll forming achieves structural cost leadership only after surpassing specific volume and batch thresholds required to amortise intensive tooling assets. In contrast, profile bending provides superior economic outcomes for high-variance and low-volume portfolios by reducing inventory requirements and eliminating changeover costs.

The model demonstrates that incorporating inventory carrying costs shifts the optimal batch sizes away from those that minimise production costs alone, highlighting the importance of holistic financial modelling in operations strategy. In most analysed scenarios, medium-level equipment configurations provide the most balanced and resilient cost-performance.

The findings can support a structured decision framework for industrial managers, demonstrating that the optimal manufacturing strategy frequently transitions from a binary technology selection to a hybrid operational deployment. Specifically, roll forming is utilised to address standardised high-volume segments, while profile bending capacity is maintained for agile, customised variants. To further provide understanding for the economic principles behind this reasoning, the study begins by setting the industrial context of the technology and the theoretical background of Economies of Scale and Scope.

Subsequent research may develop the modelling by assessing effects of various market strategies and evolving demand structures on capacity, investments, and the optimal distribution between roll forming and profile bending. Such analysis would enhance the robustness of long-term production planning.

Keywords: Comparative Cost Model, Break-Even Analysis**,** Sheet Metal Profile Manufacturing, Roll Forming, Profile Bending, Steel Forming and Shaping Fabrication, Production Strategy, Economic Order Quantity, Investment Configurations, Economies of Scale and Scope.

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List of abbreviations

AC	Average Cost
AFC	Average Fixed Cost
CapEX	Capital Expenditure
CBA	Cost Breakdown Analysis
CNC	Computer Numerical Control
CODP	Customer Order Decoupling Point
CRS	Constant Returns to Scale
DML	Dedicated Manufacturing Line
DSC	Degree of Economies of Scope
EOQ	Economic Order Quantity
FMS	Flexible manufacturing system
IRS	Increasing Returns to Scale
FC	Fixed Cost
LCC	Life Cycle Cost
LRAC	Long Run Average Cost
MES	Minimum Efficient Scale
MTO	Make-to-Order
MTS	Make-to-Stock
OpEX	Operating Expenses
PB	Profile Bending process
RF	Roll Forming process
RMS	Reconfigurable Manufacturing Systems
SC	Set-up cost
SRAC	Short Run Average Cost
TC	Total Cost
TCO	Total Cost of Ownership
TP	Total Productivity
VC	Variable Cost
WACC	Weighted Average Cost of Capital

1. Introduction

As global competition intensifies and customer demand for both low prices and product variety increases, manufacturing companies face increasingly complex decisions regarding the choice of production processes (Naldi et al., 2025). Industrial strategies that were previously stable over time are now challenged by swift technological development, shorter product life cycles and increased demands for flexibility. Against this background, understanding how different production processes and manufacturing technologies affect cost structures and competitiveness becomes a central question for companies that want to optimise their resources and make long-term sustainable investment decisions.

In the field of industrial economics and the theory of the firm, the strategic configuration of production systems is fundamentally rooted in the concepts of economies of scale (size) and economies of scope (co-production). Understanding how output volume and product diversification interact to dictate a firm's Long-Run Average Cost (LRAC) curve continues to be a foundational research question within microeconomic theory (Panzar & Willig, 1981; Stigler, 1958).

The economic investigation into the concept of scale explores how increasing the volume of a single output diminishes unit costs by amortising fixed, capital-intensive asset investments over a larger production base. The core research problem lies in identifying the optimal volume of production where marginal efficiencies are maximised before encountering diseconomies of scale, such as idle capacity, operational friction, or excessive inventory holding penalties.

On the other hand, the question of co-production-benefits, conceptualised as scope, addresses the operational advantages gained when the joint total cost of manufacturing a varied portfolio of products is lower than the aggregate cost of producing them independently. This theoretical dimension focuses on how shared inputs, flexible manufacturing technologies, and minimized changeover friction can absorb product variety without inducing severe economic penalties and therefore provide market value.

In industrial management, these two economic forces represent a classic strategic paradox. Technologies optimised for scale often require rigid standardisation, while those optimised for scope sacrifice cycle-time for agility. Mathematically modelling the precise economic thresholds, or break-even points, where these two forces intersect at varying levels of capital investment is therefore a critical research challenge necessary for guiding corporate manufacturing strategy to reduce inefficiencies and opportunity cost.

These theories are particularly relevant when comparing production processes that differ in degree of automation, flexibility and investment needs. In this thesis, two different production processes that are commonly used to make sheet metal profiles are studied. See Figure 1 for examples of these products. Establishing a precise cost calculation methodology for sheet metal forming processes is also an absolute prerequisite intended for optimising industrial logistics and material flows (Janovska et al., 2022).

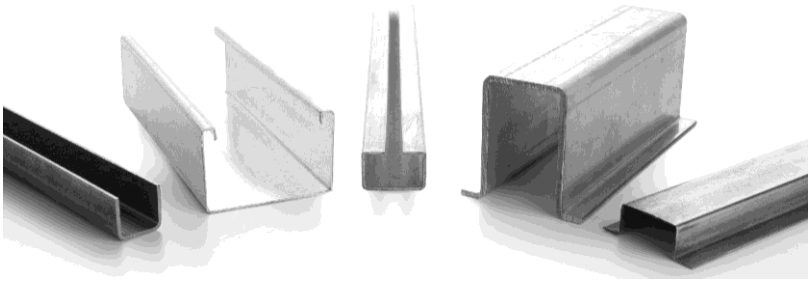


Figure 1. Examples of sheet metal profile products, either roll formed or bended.

2. Industrial context

This chapter sets the industrial context and provides a technological background information to the area of sheet metal profile manufacturing, describing the economic drivers of the two production methods as well as three distinct levels of varying technical performance, automation and investment within each.

2.1. Roll forming process

Roll forming (RF) is a continuous forming process in which the undeformed sheet metal is successively formed through a series of forming steps, using tools that shape the sheet into a finished profile (see Figure 2). By having each station perform only a small part of the total deformation, the process can be controlled very carefully, minimising unwanted deformation and guaranteeing dimensional stability (Halmos, 2005). The technology is especially suitable for long series and high production speed.

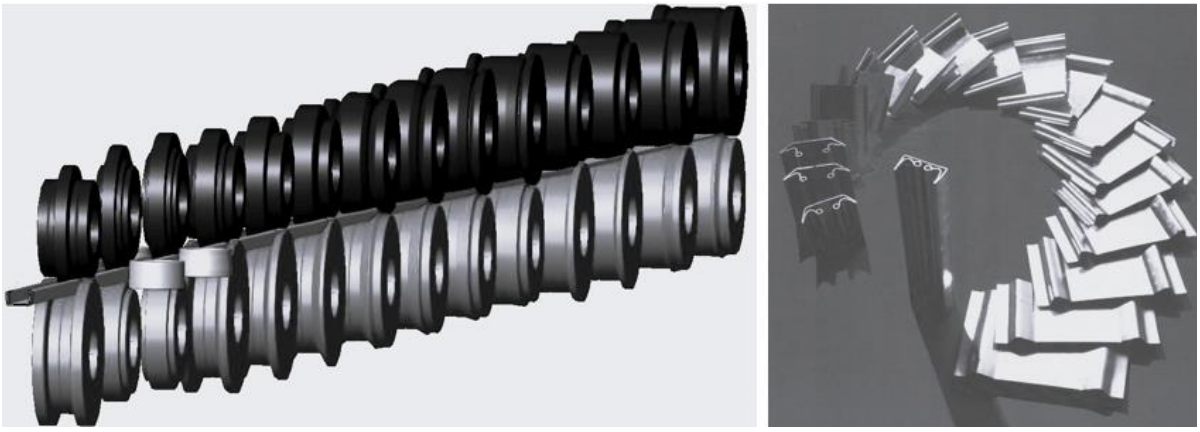


Figure 2. Left: Tooling in every forming step. Right: Shape of the profile in every forming step.

An advantage of RF is its high material flexibility. All metals that are bendable can also be roll formed, including mild steels, high-strength steels, stainless steels, aluminium and copper. In addition, the material can be pre-painted or coated, such as a zinc coating, even before forming, which can eliminate the need for expensive subsequent surface treatment. Production speeds between 10 and 40 meters per minute are common (Lindgren, 2009). In addition, profiles can be formed in a wide thickness range, which makes the technology very well suited for volume production in several industrial sectors.

A key strength is the ability to integrate multiple secondary processes directly into the RF line. Functions such as punching, riveting, welding, and more can be performed simultaneously on the RF line, meaning the profile is ready to be sold directly or passed on for final product assembly. This reduces the requirement for separate machines and complementary stations, thus creating a rational and resource-efficient production flow. The combination of material flexibility, high speed and integrated operational functions makes RF attractive in everything from the construction industry, such as profiles for roofs, walls and installation, to logistics halls and warehouse systems, white goods and more (Lindgren et al., 2015).

The economic benefits derive directly from these technical characteristics. The high production rate means that the unit cost drops quickly once the volume of production increases, making the process particularly profitable in series production. In addition, the relatively expensive tools have a long service

life as the process is based on cold forming and rolling contact. The tools are exposed to little friction and limited wear, which means that the cost of the tools is distributed over very large volumes.

The fact that several process steps can be carried out in one and the same line also enables significant savings through reduced material handling, fewer internal transports and lower labour costs. Since the profiles are in many cases completely ready for use after forming, the need for intermediate storage and post-processing is also reduced.

Finally, roll forming contributes to lower material costs because the process generates very little waste. This means that the amount of scrap is kept to a minimum, which is especially significant since raw materials typically represent one of the largest cost items in production. Personnel costs per meter of profile produced are also very low, as one and the same operator can often monitor one to two production lines, depending on the degree of automation, see Figure 3 for a typical RF line. The high degree of automation means that the working time per meter of produced profile is significantly lower than in processes that require manual handling in several steps.

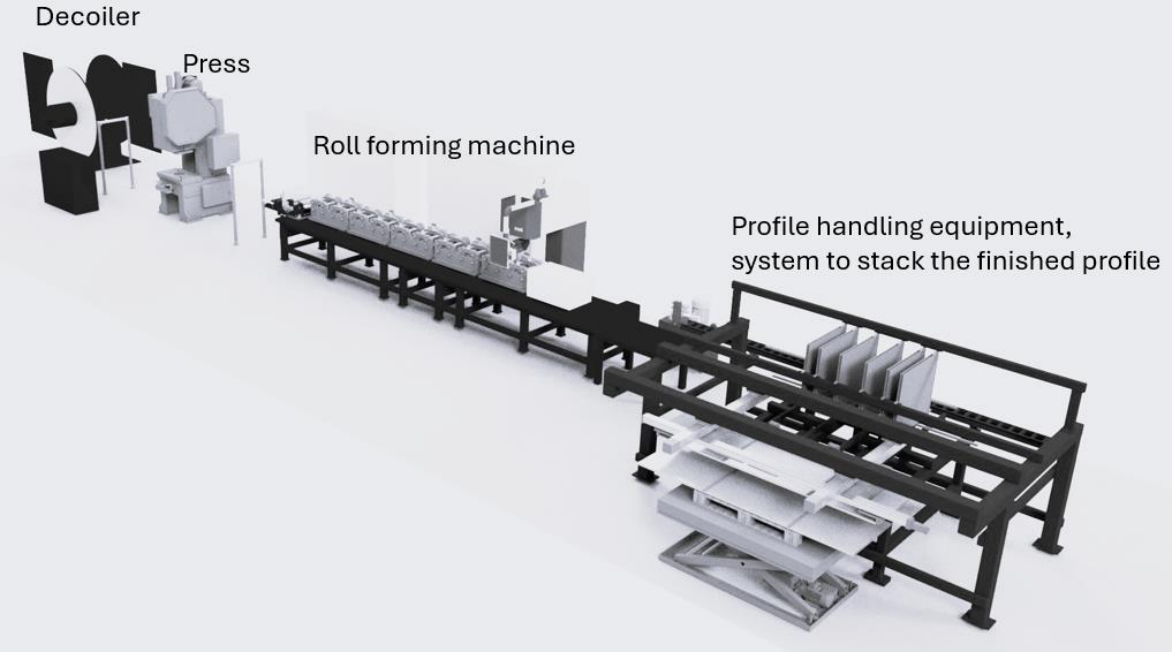


Figure 3. Layout of a typical roll forming line, from decoiler to the material handling system.

Roll forming is also closely associated with the economic concept of economies of scale, which means that the unit cost decreases as the volume produced increases. Since the process is continuous, very fast and based on large production volumes with limited manual effort, the costs are distributed over a very large number of meters produced. The high initial tooling investments are thus not a disadvantage, but rather an advantage, as they provide a very low cost per unit as volumes increase. All in all, this makes RF an exceptionally cost-effective method in situations where large batches, high productivity, and stable quality are critical to profitability.

However, roll forming is less advantageous from an economy of scope perspective. Each profile requires a specific tool package and often a large number of forming steps, making product changes time-consuming and costly. Without special investments, changeover times can amount to days, which limits the ability to switch between different products and thus reduces the flexibility of the process. This differs from, for example, profile bending, where the same machine can produce many different geometries with a short set-up time. In addition, the high tooling costs of RF mean that each new product

requires a significant initial investment, making broadening the product portfolio less attractive if volumes are not large enough.

To improve the economy of scope of roll forming, there are technical solutions, such as modular cassette systems. These make it possible to change entire tool packages in a short time, often 10-30 times faster than traditional changeovers, where the production is stopped and the tool changeover occurs in the machine line (Halmos, 2005). The cassette system allows the same machine bed to be used, but new tools can be lifted onto the machine bed in a short time, see Figure 4. This reduces both set-up time and the amount of scrap that arises during running-in, while reducing the demands on the operator's specialist expertise. Furthermore, equipment such as a double decoiler, machines that can adjust automatically and integrated processes such as punching and welding can also help increase flexibility by bringing multiple operations together in the same line. However, despite these improvements, RF is still a process optimised for high volumes rather than wide product offerings.

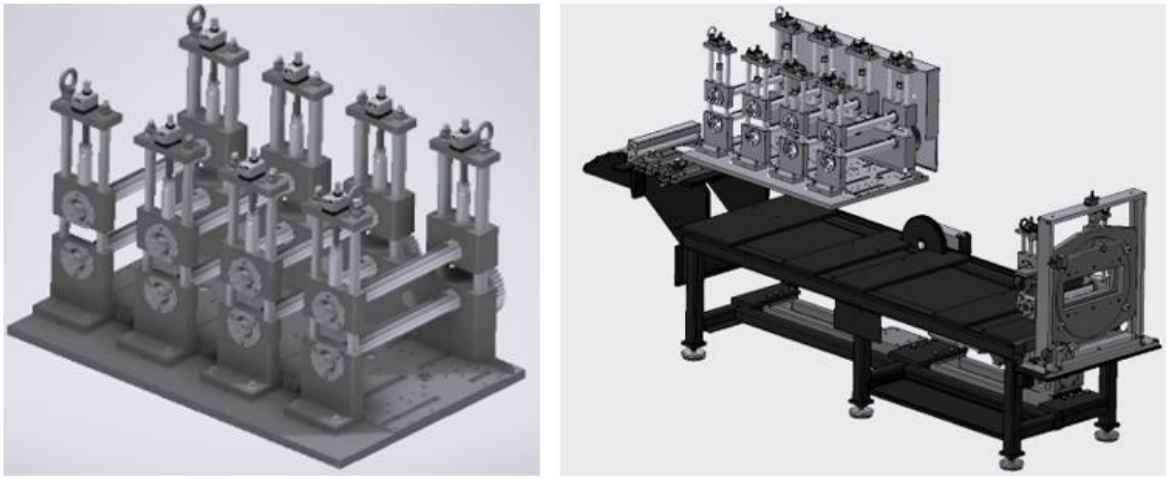


Figure 4. Left: roll forming cassette. Right: machine bed together with a cassette.

To summarise, this means that RF is a technique in which economies of scale dominate and where economies of scope can only be achieved through significant investments in automation. For companies, this means a strategic trade-off: RF is very cost-effective when volumes are high and products are relatively stable over time, though less suitable for businesses that need quick production changeovers or the ability to handle many product variants. Therefore, the choice of process becomes not only a technical issue, but rather a central part of the company's production strategy and business model.

2.1.1. Fixed and variable costs in roll forming

In roll forming processes, three central cost components arise: fixed costs (FC), set-up costs (SC) and variable costs (VC). The fixed cost consists mainly of the investment in production equipment, such as the decoiler, punch, RF machine, profile handling equipment and the tools specific to the profile to be manufactured, see Figure 2 and Figure 3. These investments are capital-intensive and are characterised by a long lifespan, which means that they must be distributed over the total volume of production in order to provide a reasonable return on the capital invested. In economic terms, this means that the fixed cost per unit of output decreases as volume (Q) increases, which is a fundamental mechanism behind economies of scale.

In addition to the fixed cost, there may arise significant costs due to setup time when switching between products, since RF requires tool changes and modifications to guarantee that each profile fulfils its

specifications. Time required for these conversions results in downtime and thus indirect costs, as personnel and equipment are stopped and not generating output. The setup time cost is particularly relevant in economies of scale analysis, as companies that produce multiple products can be negatively affected by frequent changes, while a wider product portfolio often can provide strategic benefits.

The variable costs are those that directly result from the production of each profile. The costs are personnel costs, material costs, energy consumption and the part of the set-up time, such as changing coils on the decoiler, that can be considered proportional to the production volume. These costs vary with the output level and thus affect the unit cost linearly. In combination with the fixed costs and the set-up costs, they form the total cost (TC) structure for roll forming, where the unit cost can be expressed as the sum of the variable cost per unit and the allocated fixed cost.

Overall, this cost structure means that roll forming is well-suited for the analysis of economies of scale and economies of scope advantages. High fixed costs and relatively low variable costs make the process sensitive to volume variations, while set-up time costs create a clear link between production mix and cost efficiency. This makes RF an illustrative example of how production systems can be optimised by balancing volume, product variety, and investment level.

2.1.2. Three levels of roll forming equipment

Differences in equipment level and automation create distinct cost structures in roll forming, which can be illustrated through three cases: a Small, Medium, and Large RF line. These cases differ in terms of fixed cost, set-up cost and variable cost, and together they demonstrate how investments in automation and roll forming equipment such as cassette systems, see Figure 4, shift the balance between economies of scale and economies of scope.

In the Small case, the RF line is equipped with basic machinery, manual settings, a single decoiler and traditional tool assembling directly in the RF machine. The RF machine is often a so-called start-and-stop machine, where the machine stops when cutting the profile length. The speed is between 5 and 10 meters per minute, depending on the length of the profile. This configuration results in low fixed costs, as the investment level is limited to indispensable equipment. However, the absence of a cassette system leads to high set-up costs, since each product change requires extensive manual work, long production stoppages and a considerable amount of scrap during set-up. The variable cost is also relatively high because of slower production speeds, higher labour input per meter produced and greater dependence on operator expertise. This type of line is typically suited for companies with stable product portfolios, low volume and long production runs, where changeovers are infrequent.

The Medium case represents a higher level of equipment, where cassettes, cassettes used in the machine and cassettes that the operator changes tools outside the machine, can be used to minimise the time for set-up. This means that tool change is moving from within the machine line to outside the machine. Nevertheless, a trained operator is needed to start up production, and the scrap level will be high. The fixed cost will increase compared to the small case, but the additional investment yields a substantial reduction in set-up costs. At the same time, variable costs decrease due to increased production speeds, because of investment in flying cutting equipment, which means a production speed of 10-40 meters per minute. This configuration is often chosen by companies that manage a moderate product mix and require a balance between productivity and flexibility.

In the Large case, the RF line is supplied with advanced automation and full cassette systems for all products. Time to exchange the product in the machine, which means changing cassettes, is within hours rather than days. The scrap level is minimised because no tool change is needed. These investments reduce set-up costs to a minimum, often by a factor of 10–30 in comparison to traditional changeovers. Additional features such as double decoilers, automatic coil change or strip welding equipment, high-speed punching equipment, fully automatic material handling equipment that stacks the finished profiles

and flying cutting equipment are used. Furthermore, secondary operations are integrated (e.g., welding, clinching, punching), thereby increase fixed costs. The variable cost per meter is also the lowest among the three cases, as automation reduces labour requirements, stabilises quality and reduces waste. This configuration is optimised for high-volume production with frequent product changes, where both economies of scale and improved economies of scope are strategically important.

2.2. Profile bending process

Profile bending (PB) is a step-by-step sheet metal forming process in which a flat sheet of metal is clamped in a fixed position, and a set of bending tools is moved and controlled along a trajectory that deforms the material into the desired shape. By repositioning the bending tools and rotating or flipping the material, more complex shapes can be created (Bodzás & Szanyi, 2025). The definition of "profile-" as opposed to, e.g. "box-bending" is that bending deformation coincides with only one axis instead of two or three. The paths of the tools, speed, force and bending order are usually programmed into the machine beforehand for an automatic, repeatable sequence. Bending tools can usually be optimised to achieve specific profile shapes and tolerances, and they can also be exchanged over time for maintenance or replacement.

Different materials such as mild steels, high-strength stainless steel, aluminium and copper, as well as the thickness of the sheet metal, usually require compensation factors in the machine programming to account for variances in bending stiffness, location of the neutral axis and spring back. Pre-painted or coated materials may need a protective film or special handling equipment to make sure that no scratches or marks from the bending tools occur during processing. The cycle time is highly dependent on the machine, the automation level and the programming, in addition to obviously the complexity of the part, but the changing over time between parts that are widely different in shape is essentially non-existent, if the tooling remains the same. This allows for high flexibility and seamless manufacture of "assembly kits" consisting of multiple different parts for Just-in-time (JIT) delivery rather than Make-to-stock (MTS), which promotes economies of scope and lean production ideology.

The variation of different geometries of profiles is nearly unlimited; however, one usual concern is the bending sequence and depth of the profile to allow for the tools to be able to reach down into, and be removed from, the profile. Another important factor in PB is the requirement to clamp or hold the part fixed during the forming process. Examples of profiles can be seen in Figure 1.

There are two main machine processes for profile bending, as illustrated in Figure 5. The first is called "press braking" and employs simple linear movement of an upper punch tool into a fixed bottom die that is usually V-shaped. A back gauge usually positions the sheet metal, and several parameters control the bend, such as press force, stroke length and tool shape. A human machine operator or an industrial robot manipulates the part so that the correct geometry is achieved, including flipping it over, if required.

The second method, called simply "folding" or "bending", more advanced and usually requires three or more tools, all moving independently. Clamping tools position and hold the sheet metal, whilst bending tools press the material in a rotational manner to the desired angle. Usually, the bending tools can move both up and down, eliminating the need for flipping the part over to make bends in different directions. This method is more suitable for a high automation level compared to press breaking, as it is usually quicker and more flexible to create complex shapes, but at a higher investment and setup cost.

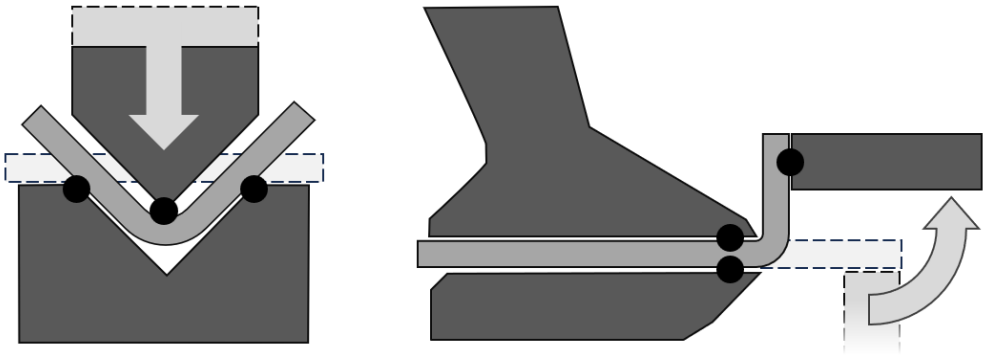


Figure 5. Left: Press braking tooling principle. Right: Folding machine tooling principle.

The financial benefits of profile bending and press braking are mainly attributed to functional versatility and low initial investment, which render these processes highly profitable for small to medium-sized batches and customised production. In contrast to nonstop processes, press braking typically utilises standardised tooling; a single set of universal V-dies and punches can generate a wide range of shapes and geometric angles. Consequently, tooling costs are reduced and distributed across multiple projects and product types, rather than being confined to a single profile design.

Rapid changeovers of tooling and overall setup enable high flexibility in production flow with minimal internal transports, resulting in reduced lead times. Preprogrammed automated tool changes and CNC adjustments minimise the need for extensive trial runs and scrap. These characteristics make the process particularly suitable for Just-In-Time manufacturing, where the capacity to produce specific parts on demand outweighs the importance of maximum production speed to stock (Janovska et al., 2022). Additionally, modern bending machines are often equipped with high-precision gauges and sensors that achieve a very high accuracy from the initial bend, accounting for material tolerances (Stefanovska & Pepelnjak, 2022).

PB and press braking are closely linked to economies of scope, wherein the average production cost declines as product variety increases. Because these machines are not restricted to a single product line, they can accept a diverse range of parts without involving considerable capital investment for each new design. Such flexibility provides a considerable benefit in markets characterised by short product life cycles or high demand for customisation. Although labour costs per unit may exceed those of fully automated rolling lines, due to the need for operator involvement, these costs are offset by the absence of expensive, product-specific tooling investments.

However, these processes offer fewer economies of scale advantages as production rates are constrained by the manual or robotic handling required for each part, resulting in relatively stable unit costs regardless of batch size. In high-volume production scenarios, the cycle time of a press brake is unable to match the continuous output achieved by RF lines. Furthermore, the physical dimensions of parts are often restricted by the length of the press bed and the capacity of the hydraulic system, which limits the production of extremely long, continuous profiles.

To improve the economies of scale in the bending, technical solutions such as robotic bending cells and automatic tool changers have been developed. These systems enable "lights-out' manufacturing, where a robot handles the material and the machine automatically swaps tools between different jobs. This reduces the personnel cost per part and increases the machine's service cycle. Furthermore, integrated software and 3D simulation can help optimise the bending sequence, limiting the reliance on highly skilled manual labour and further shrinking setup times. Despite these technical advances, press braking is still a process optimised for flexibility and variety rather than extreme mass production.

To summarise, this means that PB is a process where economies of scope dominate and where economies of scale can only be achieved through significant additional investments in automation. For companies, this means a strategic trade-off: PB is cost-effective when the product-mix is varied, and it can rapidly adapt to changes, but it is less suitable for businesses with high volume demand and low cycle time requirements. Just like for roll forming, there are a lot more than technical factors that need to be considered to make a strategic and financially optimal decision to invest in this solution.

2.2.1. Fixed and variable costs in profile bending

Profile bending and press breaking processes involve three primary cost components: fixed costs (FC), setup costs (SC), and variable costs (VC). Fixed costs primarily encompass investments in production equipment, including press brake machines, hydraulic or electric drive systems, control units, and a range of dies and punches necessary for various bend geometries. Although these investments are generally lower than those for automated roll forming lines, they remain capital-intensive and require consistent production throughput for effective amortisation. The fixed cost per unit decreases as cumulative production volume (Q) increases, serving as a key driver of economies of scale.

Set-up time costs are especially important in bending operations. Unlike roll forming, which is optimised for continuous production, press breaking usually involves more frequent changeovers. Set-up time includes selecting and mounting appropriate tooling, conducting test bends to calibrate for material spring-back, and updating the CNC program for each specific profile. These activities result in machine idle time, representing an indirect cost. Nevertheless, the high flexibility of press breaking for custom geometries means that set-up costs are often considered a necessary investment to achieve economies of scope, enabling manufacturers to accommodate high product variety without requiring dedicated line reconfigurations.

Variable costs are the expenses directly associated with producing each individual part. In press breaking, labour costs often predominate due to the frequent need for manual handling or operator intervention during part positioning. Additional variable costs include materials, energy consumption, and tool wear. As press breaking is generally more operator-intensive than roll forming, variable costs per unit are typically higher.

This cost structure demonstrates that PB and press breaking are ideal for agile, small to medium batch production. Although these processes do not achieve the high-volume efficiency of roll forming, their lower initial fixed costs and greater flexibility make them more desirable in settings where product variety and rapid design changes are prioritised over high-volume mass production.

2.2.2. Three levels of profile bending equipment

Variations in equipment functionality and automation level result in different cost structures for profile bending, as for roll forming. Three distinct scenarios are used to illustrate this: manual press breaking, a semi-automated PB cell, and a highly automated and integrated punching and bending line. These scenarios differ in fixed, set-up, and variable costs, showing how investments in CNC control and robotic handling alter the relationship between economies of scale and economies of scope.

In the manual press braking scenario (Small), production utilises a standard press brake machine, see Figure 6. This investment brings relatively low fixed costs, limited to the basic machine and standard universal tools. However, this manual process results in high setup costs for each new profile which may require the operator to load different tool segments, adjust the gauging, and conduct multiple test runs, which generate significant amounts of scrap material. Variable costs remain high due to the labour-intensive nature of the process, as the operator must support and rotate the part for each bend. Production speed is slow, and quality is highly dependent on operator fatigue and expertise. This configuration is

most appropriate for prototyping or very low-volume production, where the low capital requirement compensates for operational inefficiency.

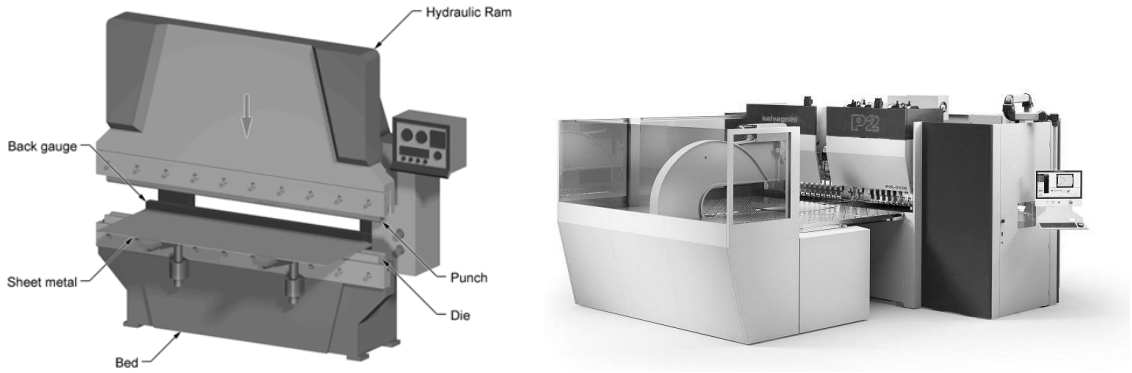


Figure 6. Press breaking and folding machines.

The second PB scenario (Medium) involves the use of an automated machine cell, in which the sheet is clamped flat while folding beams swing up and down to form flanges in both directions, whilst successively repositioning part after each bend, see Figure 6. Fixed cost increases, but this investment substantially reduces the set-up costs. The machine incorporates segmented tooling, which can be rapidly adjusted. An operator is still required to load material, but variable costs decrease because the machine supports the part's weight, allowing faster cycle times as well as improved ergonomics. This setup is typically selected by companies with a moderate product mix that require higher tolerance and faster changeovers than manual press brakes can offer.

In the third scenario (Large), the system is fully integrated and automated, see Figure 7. The machine features automatic tool changers that adjust the configuration within seconds. Fixed costs are highest, as the production line includes automatic material loading, integrated shearing and punching stations and optimised orientation and flow of material before, during and after bending. Scrap is nearly eliminated because the CNC system employs precision sensors to adjust for material variations in real time. The variable cost is continuously low, as the system can operate unattended, assuring consistent quality and maximising throughput. This configuration is designed for high-volume production environments with frequent product changes, where attaining both scale and scope is essential.

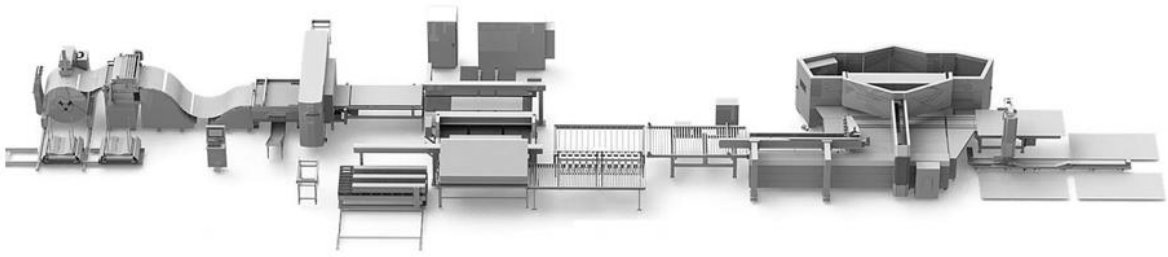


Figure 7. The layout of a typical fully automated folding machine line.

3. Theoretical Background

This chapter provides an overview of literature concerning cost structures and production decision-making within manufacturing. It includes foundational studies from a historical perspective and examines key cost concepts, such as average cost, total cost, opportunity cost, and break-even analysis, as tools for evaluating and comparing production alternatives. Although economies of scale and scope are discussed to establish a general understanding of cost behaviour, the primary emphasis is on how cost structures, production volume, and flexibility affect the relative efficiency of various processes. Additionally, the section explores trade-offs between production approaches and considerations for combining or switching between processes under different conditions. This analysis establishes a basis for evaluating break-even thresholds and identifying cost-efficient production strategies in practice.

3.1. Economies of Scale

Groenewegen (1977) describes how Adam Smith viewed the division of labour as the starting point for his theory of economies of scale. This idea became the foundation of what today is referred to as economies of scale. According to Smith, three mechanisms increase productivity: time savings, enhanced professional skills, and incentives for mechanisation and technological innovation. He illustrated this with the well-known example of the pin factory. Furthermore, Groenewegen (1977) emphasises that Smith regarded the division of labour as a self-reinforcing process: increased specialisation requires larger markets, which in turn generate scale economies. Groenewegen (1977) also shows that Smith, already in the 1700s, discussed how higher volumes reduce average costs—a principle that is now central to modern industrial economics. He concludes that Smith's ideas not only established the foundation for contemporary economic theory, but also for modern concepts such as scale effects and the role of markets in technological development.

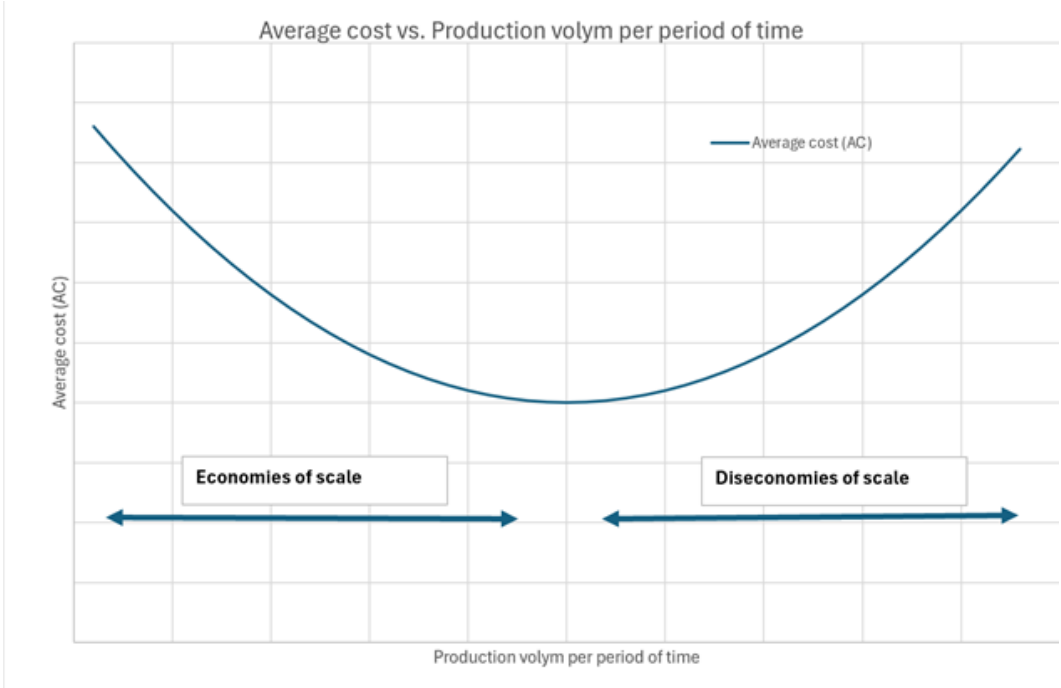


Figure 8. U-Shape curve for the average cost (AC) versus production volume per period time.

Stigler (1958) examined how firms expand their market share over time in order to determine whether an optimal firm size exists within a given market. His findings showed that a wide range of firm sizes can operate efficiently, which challenged the traditional U-shaped model of how average cost (AC)

changes with increased output, as shown in Figure 8. Instead, the empirical pattern suggested a curve that declines as volume increases, then flattens over a broad interval, and finally rises again, as shown Figure 9.

Stigler (1958) also concluded that economies of scale mainly stem from technology and plant size, but that these advantages diminish once firms become very large, leading to diseconomies of scale. This insight is one of Stigler's most important contributions: he demonstrated that diseconomies of scale arise not from technological limits but from organisational factors such as slower decision-making, increased managerial complexity, and difficulties in coordination and control.

Stigler (1958) introduced the concept of the survivor principle, which holds that firm sizes that maintain or increase their market share over time are the most efficient. A firm size that fails to survive, even if it appears theoretically ideal in terms of cost structure, cannot be considered truly efficient. A central yet sometimes overlooked aspect of the survivor principle is that it does not imply a single optimal firm size or strategy. Instead, firms may use different types of efficiencies, allowing multiple size classes to coexist within the same market. Some firms may operate at a very large scale and benefit from economies of scale, while others may remain smaller, but more flexible, specialising in niche or customised products.

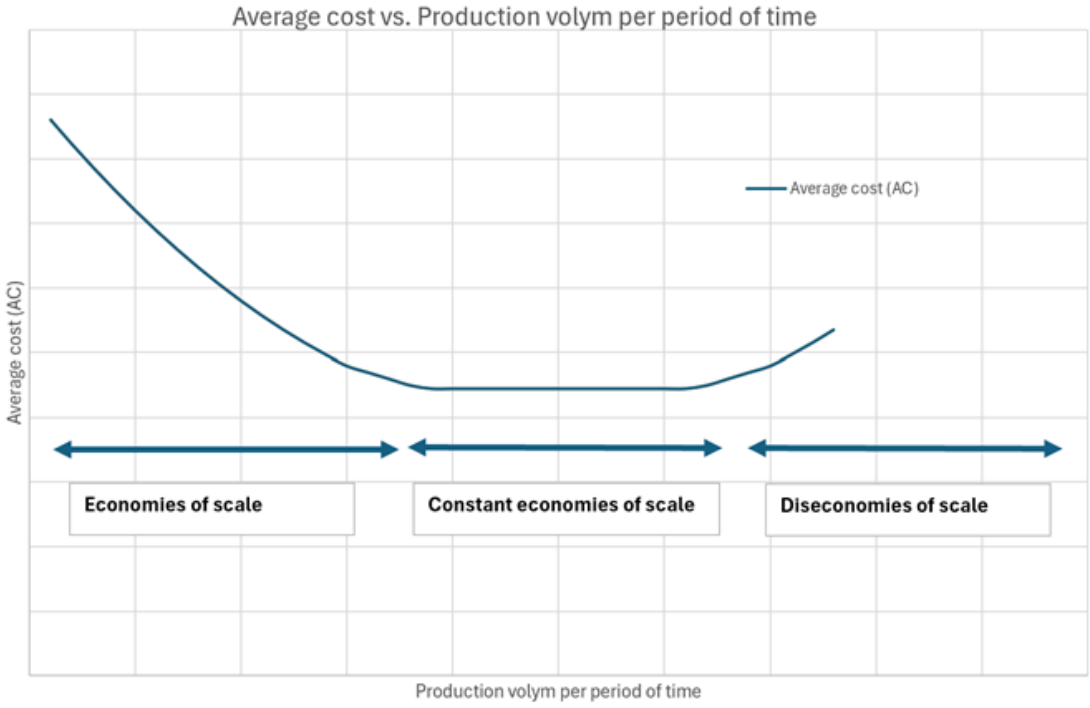


Figure 9. L-Shape curve for the average cost (AC) versus production volume per period time.

Silberston (1972) conducted extensive analyses across a wide range of manufacturing industries to examine how economies of scale arise and how average cost (AC) is affected by various factors such as production volume, product lifetime, specialisation, and vertical integration. His findings showed that most AC curves take an L-shaped form: substantial cost reductions occur early as output increases, after which the curve levels off near the Minimum Efficient Scale (MES), Figure 10. A major contribution of Silberston's (1972) work was his analysis of cost increases at 50 percent of MES, as shown in Figure 10, which provided a practical measure of the strength of economies of scale. By quantifying how costs behaved at this point, he offered a more empirically grounded understanding of scale advantages and their limits.

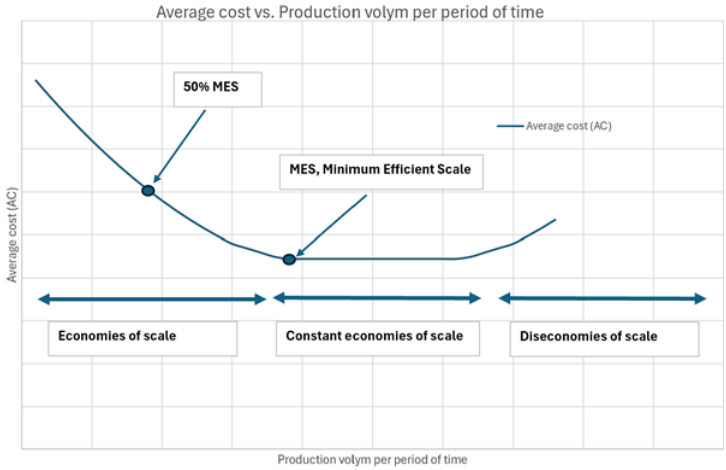


Figure 10. L-Shape curve for the average cost (AC) and the MES, Minimum Efficient Scale.

For this, Silberston (1972) used the mathematical function *Equation 1 - 4* to describe the relationship between the economies of scale and the average cost.

$$AC(Q) = aQ^n \qquad \textit{Equation 1}$$

Where:

$$AC(Q) = average\ cost\ of\ production$$
$$Q = quantity$$
$$a = constant$$
$$n = scale\ parameter\ (negative\ value\ for\ economical\ of\ scale)$$

The cost increase at 50% MES, shown in Figure 10, is defined as:

$$\Delta = \frac{AC(0.5MES) - AC(MES)}{AC(MES)} \times 100\% \qquad \textit{Equation 2}$$

Equation 1 and Equation 2 together give:

$$\Delta = (0.5^n - 1) \times 100\% \qquad \textit{Equation 3}$$

Solve for *n* in Equation 3 gives:

$$n = \frac{ln\left(\frac{AC(0.5MES)}{AC(MES)}\right)}{ln\ (0.5)} \qquad \textit{Equation 4}$$

Silberston (1972) showed, according to the study, that with the help of *Equation 4,* one can calculate if there is a possibility to get the economies of scale if you increase the volume. Table 1 presents the interpretation of *n* at different values.

Table 1. Interpretation of the value of the n parameter.

Value of n	Scale effect	Interpreting for economics of scale
n = 0	No scale effect	AC = constant
-0.05 > n > -0.15	Weak scale effect	AC = Slightly declining against MES
-0.15 > n > -0.30	Moderate scale effect	Clear cost fall – MES important
-0.30 > n > -0.60	Significant scale effect	Capital intensive industry requires high volume.

Kim (1987) examined firms producing more than one product and analysed how this affects economies of scale. His study focused on a water supplier that provided water to both households and businesses. The results showed that large waterworks exhibit approximately constant returns to scale, while small waterworks display increasing returns to scale. He also identified differences between household and industrial water: household water supply showed increasing marginal costs and diseconomies of scale, whereas water supplied to businesses demonstrated the opposite pattern, with clear economies of scale.

Based on these findings, Kim (1987) concluded that not only a firm's product mix but also its customer mix influences economies of scale. For the waterworks, this meant that two factors shape scale economies: production volume and the composition of the customers. His overall conclusion was that multiple firm sizes are able to coexist within the same market, depending on their respective product and customer mix.

Besanko et al. (2017) note that although the traditional cost curve is typically depicted as U-shaped, empirical evidence instead suggests an L-shaped pattern. In this form, average cost declines rapidly as output increases and then levels off near the Minimum Efficient Scale (MES). According to Besanko et al. (2017), the decline in the average cost with rising production volume is caused by several mechanisms: the spreading of fixed costs, increased specialisation, inventory related effects, and technical factors such as the cube-square rule. In addition, the authors emphasize the role of learning curve effects, whereby costs fall as firms accumulate experience, improve their processes and improve efficiency over time. These learning effects further reinforce the cost advantages enjoyed by firms with high production volumes.

Swann (2009) argues that process innovations are central to firms' ability to achieve economies of scale. Because process innovations aim to reduce production costs through more efficient methods or new technologies, they become particularly valuable in industries where large production volumes already generate cost advantages. Swann (2009) emphasises that such innovations are often expensive to develop and implement in high-volume industries, which gives an advantage to firms that are already large. Once a process innovation is adopted, it can further reduce marginal costs and thereby reinforce existing economies of scale. In this way, process innovations not just enhance efficiency, but also strengthen the competitive position of firms capable of operating at a substantial scale.

Keat et al. (2014) explain that economies of scale arise when the long-run average cost (LRAC) decreases as output increases, reflecting that resources can be utilised more efficiently at higher production volumes. In the long run, all inputs, both capital and labour are variable, which allows firms to adjust plant size, production technology, and organisational structure to achieve the most cost-efficient scale of operation. However, the authors also emphasise that economies of scale are not unlimited. When a firm grows too large, diseconomies of scale may emerge, for example, through rising coordination and administrative costs. An important implication is that firms seek to reach their Minimum Efficient Scale (MES), the level of output at which LRAC is minimised, and economies of scale are fully realised.

While LRAC captures the cost-minimising possibilities available when all inputs can be adjusted, short-run cost conditions also have a crucial role in understanding how firms move toward their efficient scale. In the short run, capital is fixed and the firms are constrained by their existing plant size, infrastructure, warehouse capacity, and maybe also machinery while only labour can be varied. This gives rise to a series of short-run average cost curves (SRAC), each corresponding to a specific level of fixed capital, shown in Figure 11. As output expands within a given plant size, the firm may initially experience declining SRAC due to better utilisation of its fixed resources, but eventually rising SRAC as diminishing marginal returns to labour set in.

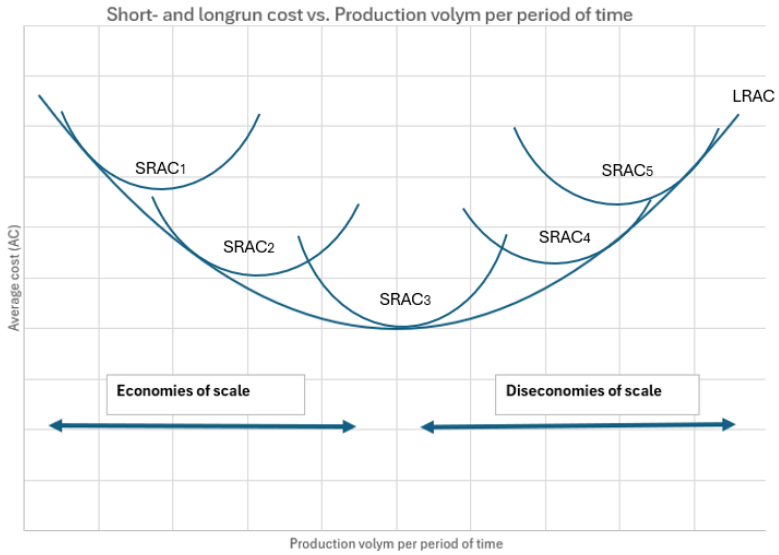


Figure 11. Short - and long run average cost versus production volume per period of time.

The relationship between SRAC and LRAC is therefore crucial to understanding economies of scale. The LRAC curve can be viewed as an "envelope" of the firm's SRAC curves: it traces the lowest attainable average cost for each output level, if the firm can freely adjust all inputs. In a typical diagram such as the one you mention, where both LRAC and SRAC are shown, Figure 11, the LRAC curve is tangent to each SRAC curve at the output level where that particular manufacturing plant size is optimally utilised. This illustrates how firms transition from short run constraints to long run efficiency and why understanding both cost perspectives remains essential for strategic decisions concerning capacity, production planning and market structure.

3.2. Economies of Scope

While the concept of economies of scale strives to achieve efficiency through volume, a strategy which has dominated manufacturing since the industrial revolution, an alternative approach, coined economies of scope, emerged later in the 20th century. It represents a shift towards understanding how variety and range in a firm's product offering can drive a competitive advantage (Besanko et al., (2017).

The term "economies of scope" was popularised by Panzar & Willig (1981) to describe cost savings resulting from the scope rather than the scale of an enterprise. They defined a production process as exhibiting scope economies when it is less costly to combine two or more product lines in one firm than to produce them separately in specialised units (Panzar & Willig, 1981).

Historically, this observation predates the term itself, and early scholars such as Clark (1925) suggested that the origins of the multiproduct firm sprang from the opportunity to exploit excess capacity. They argued that firms diversify when they possess inputs, physical or intangible, that are not limited by a

single product line. Keat et al. (2014) elaborate on this by noting that a firm's expansion into different lines of business naturally increases its scale of operation, linking scope closely to scale.

The technical core of economies of scope is the concept of "subadditivity" in the cost function, which holds that it is cheaper for a single firm to produce a given level of total output than for two or more firms to produce that same total output separately. Panzar & Willig (1981) provided the definitive mathematical expression that explains that economies of scope exist if the total cost of producing quantities of two different products is lower when produced by a single firm than the sum of the costs of producing them in separate firms:

$$TC(Qx, Qy) < TC(Qx, 0) + TC(0, Qy) \qquad \textit{Equation 5}$$

Where:

$$TC(Q) = toltal\ cost\ of\ production$$
$$Q = quantity$$
$$x = first\ product$$
$$y = second\ product$$

This formula captures the inherent efficiency of joint production. Besanko et al. (2017) offer an alternative interpretation focusing on incremental cost, where the cost of adding products to an existing production is lower than the cost of producing from scratch in a standalone facility, with this formula:

$$TC(Qx, Qy) - TC(0, Qy) < TC(Qx, 0) \qquad \textit{Equation 6}$$

The degree of economies of scope formula measures the cost efficiency that a firm gains by producing multiple products together rather than individually. It quantifies the percentage reduction in total cost when, for example, two products are produced by one firm instead of two separate firms:

$$\text{DSC} = \frac{TC(q1) + TC(q2) - TC(q1,q2)}{TC(q1,q2)} \qquad \textit{Equation 7}$$

Where:

$$DSC = Degree\ of\ Economies\ of\ Scope,\ \ (see\ \text{Table 2})$$
$$TC(q1) = Total\ Cost\ of\ producing\ only\ good1$$
$$TC(q2) = Total\ Cost\ of\ producing\ only\ good\ 2$$
$$TC(q1, q2) = Total\ cost\ of\ producing\ good\ 1\ and\ good\ 2\ together$$

Table 2. Interpretation of the degree of economies of scope.

DSC value	Favour Economies of Scope
DSC > 0	There are economies of scope. It is recommended that two firms cooperate, or systems merge, to produce together.
DSC = 0	There are no economies of scale nor economies of scope.
DSC < 0	There are diseconomies of scope. It is not recommended for the two firms work together. Diseconomies of scope means that it is more efficient for two firms to work separately since the merged cost per unit is higher than the sum of stand-alone costs

Since economies of scope broaden rather than narrow the focus of an organisation, it is like choosing whether to be a generalist rather than a specialist within a firm. While this may seem to apply more to a commercial firm, it does apply to just about any type of organisation. Figure 12, shows that the business may expand its economies of scope by introducing new but related product lines.

- With only one product line (one 'enterprise'), the cost per unit is C1
- Adding a second product line decreases costs per unit to C2
- Adding a third product line decreases costs per unit to C3

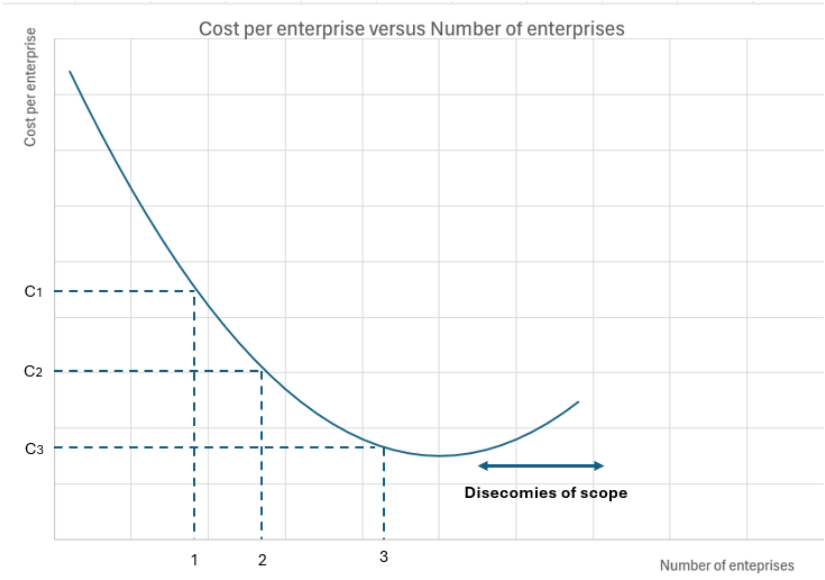


Figure 12. Cost per enterprise (product line) versus number of enterprises

At some point, the number of enterprises creates inefficiencies, and the curve begins to slope upward, indicating diseconomies of scope. Economies of scope and "strategic fit" are closely linked. It is easy enough to think of how a business can expand its scope, but another thing altogether to expand it economically, and that's what strategic fit is all about. It occurs when an organisation's activities, resources, and goals are properly aligned with its external environment, often crucial during mergers and acquisitions.

An important component is "dynamic capabilities" that are the higher-order abilities to sense, seize, and reconfigure these resources to maintain that strategic fit in volatile market conditions. A catalyst for enabling economies of scope is the efficient use of so-called "quasi-public" simply shared inputs (Panzar & Willig, 1981). Unlike private inputs that are consumed by a single product (e.g., raw materials like sheet metal), a quasi-public input is a resource that, once procured for one product line, remains available, either wholly or in part, to support others without a proportional increase in cost.

Panzar & Willig (1981) and Besanko et al. (2017) categorise these inputs into four types of assets: Physical Capital: Indivisible equipment or factory buildings that can be used for multiple processes without complete congestion. Human Capital: Specialised engineering know-how or management talent that can be applied to different manufacturing processes (e.g., roll forming and profile bending). By-products: Situations where the production of one good (e.g., wool) inevitably offers a second input (e.g., mutton) for another product line. Intangible Assets: Brand reputation, R&D databases, and organisational routines.

In the 1980s, Michael Porter (1987) transitioned the concept of scope from production theory to corporate strategy. Porter operationalised economies of scope through the Value Chain, identifying two main ways diversified firms create value: sharing activities (work) and transferring skills (knowledge). Sharing activities enhances competitive advantage by lowering costs or raising differentiation. For example, a firm can leverage a common physical distribution system and sales force for different products as well as share advertising and after-sales service activities across several product ranges, spreading the fixed costs over a wider combined base.

However, Porter (1987) also introduced a critical warning through the so-called "Better-off Test", implying that a diversified company must provide tangible advantages to its business units that they could not achieve independently. He argued that many firms will risk failure by relying on "paper synergies" (rationalising acquisitions after the fact) rather than actual operational integration. The “test” focuses on whether a new business unit and the existing corporation create synergistic value that improves their competitive position. It evaluates if the parent corporation offers skills, resources, or capabilities that give the new unit a competitive advantage, or if the new unit strengthens the parent’s existing business. The move is only justified if it brings more than just incremental growth (1+1= 2); it must lead to higher profits and a stronger market position.

Besanko et al. (2017) expand on the role of management in creating scope economies through the "dominant general management logic". This suggests that managers of diversified firms can spread their unique skills in conceptualising business, resource allocation, and organisational design across different areas. This is consistent with *Resource-Based Theory*, which posits that for a competitive advantage to be sustainable, it must be underpinned by resources that are scarce, immobile, and difficult to imitate. Managerial routines and other "well-practised patterns of activity" are key intangible assets that drive scope economies according to Besanko et al. (2017, pp. 353-355).

In the context of sheet metal fabrication, the decision to integrate roll forming (traditionally a scale-intensive process) with profile bending (a scope-intensive flexible process) constitutes a strategic choice to exploit scope economies. By using shared engineering expertise, common material procurement and integrated logistics, a firm can reduce the total cost of ownership across the entire production lifecycle.

A critical engagement with the sources reveals a tension between the production-theoretic view and the managerial view. Besanko et al. (2017) distinguish between technical efficiency (using the least-cost process) and agency efficiency (minimising coordination and transaction costs). While Panzar and Willig (1981) focus on the technical cost-reduction potential of quasi-public inputs, Porter and Besanko warn of the rising bureaucracy and influence costs. As a firm expands its scope, the central office must tame the self-interest of employees and manage the "administrative overload" that occurs when functional departments are overextended.

Solow (1956) argues that under neoclassical conditions of constant returns to scale the system eventually approaches steady growth. However, in the domain of scope, Porter (1987) argues that most diversified companies fail to add value, often selling for a "conglomerate discount" where the whole is valued less than the sum of its parts.

Keat et al. (2014) note that service productivity often resists growth because services cannot be stored and require standardisation to leverage assets. In manufacturing, however, Industry 4.0 technologies such as 3D printing and flexible robotics to enable "customisation at scale", effectively blurring the line between scale and scope.

The literature suggests that, for SMEs in the industrial sector, growth is positively associated with investments in both tangible and intangible assets. Investments in tangible assets (machinery) are essential for expansion, while intangible assets (R&D, skills) are critical for the capitalisation of those investments into EBITDA growth.

3.3. Connection to Manufacturing Strategy

Choosing between economies of scale and scope involves balancing the benefits of volume against those of variety. Strategically choosing the right path requires an analysis of market demand, production technology, and the firm's internal coordination capabilities. The primary trade-off is between standardisation (Scale) and variety (Scope), see Table 3.

Since economies of scale benefit from achieving a cost advantage by spreading indivisible fixed costs (such as specialised machinery or R&D) over a high volume of a single product, they require high throughput. If demand is insufficient, the firm faces high average costs from underutilised capacity. Furthermore, scale-intensive firms are often less agile and can be vulnerable to disruptive technologies.

In economies of scope, the cost savings are achieved by producing two or more goods jointly rather than separately. This is driven by "quasi-public inputs", i.e. resources like a factory building, engineering expertise, or a distribution network that can be shared across product lines without being fully consumed by one. The risk as variety increases, so do managerial complexity and coordination costs. A frequent challenge is the “compromise cost” which arises when a shared activity, such as a single sales force serving two products, operates less efficiently than specialised activities.

Table 3. Summary table where each strategy is favourable for a firm

Feature	**Favour Economies of Scale**	**Favour Economies of Scope**
Product Type	Standardized, high-volume	Differentiated, complex, or customized goods
Market Condition	Large market, stable demand	Niche markets, volatile or fragmented demand
Core Advantage	Cost leadership through volume	Flexibility and variety (differentiation)
Key Risk	Asset obsolescence	Administrative overload and "compromise costs"

To evaluate the conditions and make a strategic choice, firms can apply the following four frameworks:

Minimum Efficient Scale (MES), defined as the smallest level of output at which economies of scale are exhausted according to Besanko et al. (2017, p. 14). The economies of scale approach are favoured if market demand far exceeds the volume where economies of scale peak. Economies of scope is favoured in a niche market and where horizontal diversification allows shared assets to reach profitable volumes. The Better-Off Test (Porter, 1987, p. 5) conveys that economies of scope should be pursued only if a new unit gains a genuine competitive advantage through operational synergies (e.g., shared logistics) rather than just corporate overhead sharing, which may otherwise be negated by the cost of complexity. Quasi-Public Inputs: Diversify if you possess underutilised, immobile resources—like specialised human capital—that can be applied to new product lines at a low incremental cost. Flexible Automation: Use modern modular technology to pursue both simultaneously. If your industry allows for digital manufacturing, prioritise flexible assets that support customisation at scale.

3.4. Break-even analysis and cost comparison

Recent research highlights that break-even analysis should not assume symmetric cost structures around capacity. Zhang (2024) shows that firms face asymmetric adjustment costs when output deviates from capacity, with higher costs often associated with scaling down production compared to scaling up. This implies that average cost curves are asymmetrically U-shaped, which affects the break-even point. In the context of comparing production processes such as roll forming (scale-oriented) and profile bending (scope-oriented), differences in capacity rigidity and flexibility may lead to distinct break-even levels. Therefore, break-even analysis should account not only for fixed and variable costs, but also for capacity-related adjustment costs and flexibility differences.

Break-even analysis is commonly used to compare alternative production systems by identifying the production volume at which their total costs are equal. This intersection point provides a clear decision rule for selecting the most cost-efficient method depending on demand. Al-Kahtani et al. (2014) applied cost-benefit analysis to determine such break-even volumes between different manufacturing configurations, demonstrating how production volume influences system choice. This approach is directly applicable when evaluating trade-offs between production methods with different cost structures.

Cafferky et al. (2014) emphasize that, in addition to determining profitability thresholds, break-even analysis is particularly useful for evaluating alternative production methods by analysing how differences in fixed and variable cost structures influence total costs at different output levels. By applying the total cost method, decision makers can identify the volume at which competing alternatives yield equal total costs, thereby determining the point at which it is economically rational to switch from one production method to another. Consequently, the selection between alternatives becomes highly dependent on production volume, where methods characterised by lower fixed costs are generally preferred at lower output levels, while methods with lower variable costs become more advantageous as volume increases.

3.5. Identified research gaps

Research on roll forming and press brake bending is extensive when it comes to technical aspects such as material behaviour, FEM modelling, and process optimisation (e.g., International Journal of advanced manufacturing technology; Journal of Materials Processing Technology). However, studies that analyse these processes from a production economics perspective are largely absent. In the reviewed literature, no peer-reviewed articles were found that address cost comparisons, batch sizes, or break-even analyses between roll forming and press brake bending. As a result, the choice between the two methods in industrial practice is often based on experience or supplier estimates rather than scientifically grounded models. The lack of economically oriented research therefore represents a clear gap in the literature, which this study addresses by developing a systematic break-even analysis for the two manufacturing methods.

4. Purpose and research questions

Modern manufacturing companies are increasingly required to balance cost efficiency with a high degree of product variation driven by dynamic and globalised markets (Pine et al., 1993). Within sheet metal profile fabrication, this challenge is reflected in the choice between continuous roll forming and step-by-step profile bending, which differ in cost structures, flexibility and investment requirements, broadly reflecting scale-oriented and scope-oriented production characteristics. Roll forming is generally associated with efficient high-volume production, whereas profile bending allows greater adaptability to variation. However, recent developments in machine design and production system automation have reshaped the economic relationship between these processes, making traditional assumptions less clear-cut and increasing the need for structured approaches to evaluate their relative performance under different production conditions.

The purpose of this study is to analyse at what production volumes these processes reach break-even relative to each other through a structured cost analysis. By combining empirical production data with established cost modelling—incorporating elements such as fixed cost allocation, setup costs, flexibility costs and capital investments—the study seeks to identify the key cost drivers influencing the break-even point. Based on this, the study aims to develop a decision-support framework that enables manufacturing firms to determine the most cost-efficient process depending on volume, product mix, product complexity and investment level.

Research Question:

At what production volumes and batch sizes do alternative processes achieve break-even relative to each other, and which cost drivers explain the differences in cost performance?

5. Method

This chapter describes the research design, data collection methods, and analytical techniques used to determine and quantify the economic trade-offs between three levels of roll forming and profile bending, respectively. The methodology is constructed to enable a transparent, reproducible, and comparable analysis of three equipment levels for each process: Small, Medium, and Large. Data collection is based on both primary sources, in the form of supplier data and real-world production results, and secondary sources in the form of literature on economies of scale and scope.

5.1. Sampling

The study employed purposive (judgmental) non-probabilistic sampling, according to Ghauri (2020), to ensure the selection of "information-rich cases" relevant to the sheet metal industry. This method is designed to prioritise the depth and relevance of data over the broad statistical representativeness of random sampling. This strategy is particularly effective in qualitative or exploratory research where the goal is to obtain insights that require specialised real-world field experience relevant to the research problem, as in this study.

The sample consisted of three distinct cases:

> **Case 1:** A roll forming machine manufacturer and dealer to provide data on equipment costs.
>
> **Case 2:** A profile bending machine manufacturer and dealer to provide data on equipment costs.
>
> **Case 3:** A manufacturer of profile products (customer of both types of machines) to provide real-world operational data, including maintenance, scrap rates, and operator competence levels. Market data from this type of firm was used to construct a production scenario that includes volumes in the number of yearly products and variance in product-mix described in the next chapter.

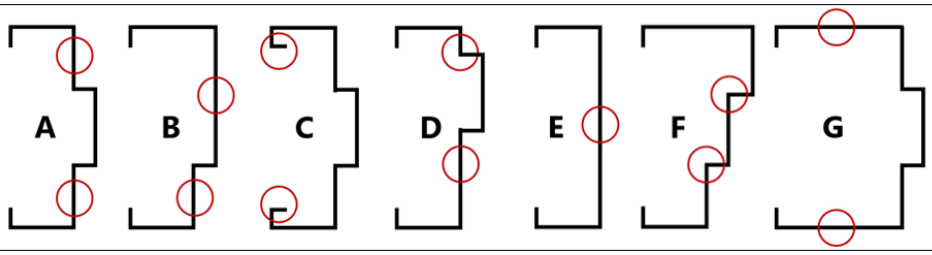
The selection ensured that both technical and practical perspectives were integrated, and that the models are not only based on theoretical assumptions but also validated against actual industrial practice.

5.2. Material

5.2.1. Product scenario

A production scenario was constructed from studying market data to provide context and the possibility to value the efficiency of each manufacturing method to fulfil a company's needs. A product mix of seven different profiles (A-G in Table 4) was chosen to represent the market variance of sheet metal frames. Each frame set has two sides and one top for a total of three cut profiles. The market share of each profile variant was researched and combined with a calculation of the number and dimensions for total yearly production volume of 100 000 frame sets consisting of 500 000 meters of profile, or about 1500 tons of steel. The variance that drives the number of profiles is primarily the interface with the frame insert, but also the connection to the wall and the depth of the profile.

Table 4. Scenario based profile types and production volumes.

Profile type									
	A	B	C	D	E	F	G		
Application	Equal egress	Single egress	Drywall	Offset	Cased Open	Doubble egress	Variable Width	**Total**	Unit
Market share	45%	30%	12%	8%	3%	1,5%	0,5%	**100%**	%
Total Sets	45 000	30 000	12 000	8 000	3 000	1 500	500	**100 000**	set
Total Profiles	135 000	90 000	36 000	24 000	9 000	4 500	1 500	**300 000**	profile
Total Length	225 000	150 000	60 000	40 000	15 000	7 500	2 500	**500 000**	meter

5.2.2. Cost of production equipment

In addition to market and production data, the study incorporates a comprehensive set of cost and equipment parameters that form the empirical basis of the cost modelling. These include machine investment costs across the three equipment configurations (Small, Medium, and Large), tooling costs, setup times and associated changeover expenses, as well as detailed information on production speeds and labour requirements. Further operational data were collected on scrap rates, material cost per meter, and energy and maintenance expenses. Financial parameters, such as the weighted average cost of capital (WACC) are also included. Together, these data inputs feed into the product cost model (chapter 5.4), enabling the construction of comparable cost structures for both production processes and allowing a consistent evaluation of their economic performance.

5.3. Procedure

The study followed a structured four-stage methodological framework to encompass data collection, scenario construction, mathematical modelling, and ethical compliance. During the initial data collection phase, primary technical figures was sourced from suppliers via product specifications, quotation documentation, and investment calculations, supplemented by semi-structured interviews with technical sales representatives and application engineers to ensure qualitative depth (Creswell & Creswell, 2018, p. 182)

Complementary quantitative data from manufacturing companies, including production logs, historical order results, scrap levels, and machine-specific operating costs, were aggregated and cross-verified to mitigate data bias and ensure consistency and comparability. Subsequently, the construction of scenarios utilised industry-standard distributions to model product mixes through tables and schematics. This empirical basis facilitated the determination of annual volumes, batch sizes, machine costs, and production schedules, aligning the model with real-world manufacturing constraints (Slack & Lewis, 2011, p. 245)

The modelling and calculations phase entailed a sequential implementation of capital and production cost models alongside the Economic Order Quantity (EOQ) framework (Ford, 1913). The mathematical progression moved from calculating hourly capital costs and variable costs per unit to calculating setup costs per batch. The study maintained rigorous ethical issues by complying with the Swedish Research Council's guidelines (2017). This included fulfilling the information requirement by educating all participants on the research objectives, securing oral consent and guaranteeing confidentiality through the anonymisation of commercially sensitive data. As the study focused exclusively on machine-level production data rather than individual personal information, the usage of data remained strictly within the confines of academic research (Swedish Research Council, 2017, p. 14).

5.4. Analysis strategy

5.4.1. Product cost model

The cost-model described in this section is used to estimate the average cost (AC) per unit when creating a profile. The model is structured around two main elements: the capital cost associated with the investment in production equipment and the running cost when profiles are produced. The capital cost is derived by combining annual depreciation with the cost of tied-up capital based on the WACC, which is then allocated across the production volume. The production cost is further divided into material, labour, and other fixed costs, enabling an analysis of how the average unit cost varies with batch size. This framework provides a transparent basis for identifying cost drivers and evaluating scale effects in the production process.

These costs increase linearly with output. Together with fixed and set-up costs, they comprise the total cost (TC) structure, where the average unit cost (AC) is calculated as follows:

$$AC = VC + \frac{FC + SC}{Q} \qquad \textit{Equation 8}$$

5.4.2. Capital cost per unit

The total capital cost is calculated by combining annual depreciation with the yearly cost of tied-up capital, based on the weighted average cost of capital (WACC), *Equation 9 - Equation 13*. To obtain a unit-level measure, the total annual capital cost is then divided by the yearly production hours for one shift, which provides the capital cost for the equipment per hour. The results are presented in Appendix B, Table 8, where three investment scenarios: Small, Medium, and Large, are compared to illustrate how different equipment investment levels influence the capital cost.

Depreciation

$$Depreciation_{per\ year} = \frac{Investment - Residual\ value}{Number\ of\ years} \qquad \textit{Equation 9}$$

Capital cost per year (WACC)

$$Average\ tied\ up\ capital = \frac{Investment + Residual\ value}{2} \qquad \textit{Equation 10}$$

$$Capital\ cost_{per\ year} = Avearge\ tied\ up\ capital * WACC \qquad \textit{Equation 11}$$

Total yearly capital cost

$$Total\ capital\ cost = Depreciation_{per\ year} + Capital\ cost_{per\ year} \qquad \textit{Equation 12}$$

Capital cost per hour

$$Capital\ cost_{per\ unit} = \frac{Total\ capital\ cost}{Production\ hours} \qquad \textit{Equation 13}$$

5.4.3. Production cost

In the analysis of production cost, the cost function $C(Q)$ is used, see *Equation 14*, which incorporates variable cost, set-up cost and the fixed costs associated with operating the production equipment. Based on this formulation, the average cost per unit is derived as a function of batch size, see *Equation 15*, in to capture how the cost structure responds to different production strategies. The resulting relationships are presented in Figure 13, where the average cost is shown for alternative investment levels in production equipment classified as Small, Medium, and Large.

To complement the cost analysis, the corresponding production time required to produce each batch is also calculated. These time estimates, which reflect differences in equipment capacity and capacity, are presented in Figure 14. Together these figures provide the basis for the following analysis of economies of scale and equipment choice, which is discussed in the results section.

$$C(Q) = C_{VC(Q)} + C_{SC(Q)} + C_{FC(Q)} \qquad \textit{Equation 14}$$

$$AC(Q) = \frac{C(Q)}{Bacth\ size} \qquad \textit{Equation 15}$$

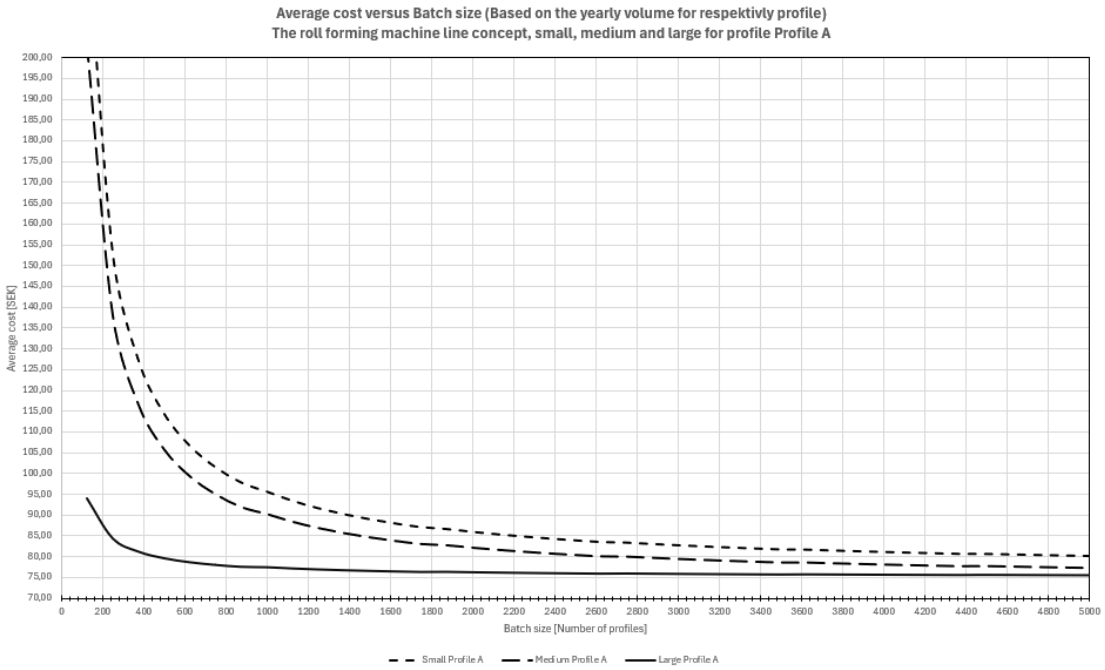


Figure 13. Average cost AC(Q) for different batch size, for respectively machine investment.

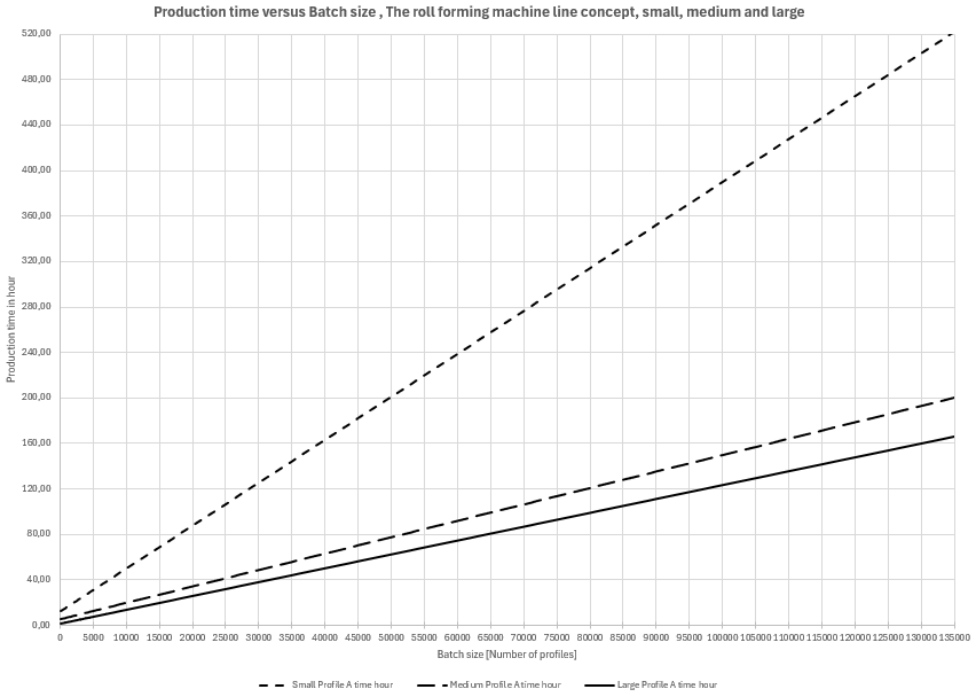


Figure 14. Production time for different batch size and respectively investment.

5.4.4. Cost of stock

In production environments where roll forming is utilised, batch sizes become a strategically important parameter. Roll forming is characterised by relatively high setup times and setup costs, as the tooling adjustments and machine configurations required for each product change are both time-consuming and resource-intensive. This creates a natural incentive to produce in larger batches to distribute the setup cost over more units. However, increasing the batch size also results in a higher average inventory level, which in turn raises the inventory carrying cost. Thus, the choice of batch size becomes an economic trade-off between minimising setup-related costs and avoiding excessive capital tied up in inventory.

The classical EOQ model (Economic Order Quantity), developed by Ford (1913), is used to determine the economically optimal order or production quantity. The model is based on a cost trade-off between two key factors: setup costs, which encourage larger production batches, and inventory holding costs, which increase with larger quantities produced or ordered. By minimising the total cost per unit, an optimal lot can be identified. However, as highlighted by Alnahhal et al. (2024), the classical EOQ model is a strong simplification of reality, since it relies on limiting assumptions such as constant demand, fixed costs and absence of supply logistics disruptions. Below and on the next page, the deduction of the mathematical equation underlying this model is shown.

Assume:

Yearly demand of profiles:	D
Number of profiles produce at same time, batch size:	Q
Set-up cost, cost of change profiles	SC
Average cost of produce a profile	AC
Cost of storing a profile (Unit cost)	*Cstock*

Set-up cost:

$$Csetup(Q) = \frac{D}{Q}SC \qquad \textit{Equation 16}$$

Cost of stock

$$Cstock(Q) = \frac{Q}{2}Cstock, \qquad where\ \frac{Q}{2}\ is\ the\ average\ stock \qquad \textit{Equation 17}$$

Cost for producing all profiles (assume this cost is constant, independent of batch size)

$$Cprod = D * AC \qquad \textit{Equation 18}$$

Total Cost

$$Ctotal(Q) = Csetup(Q) + Cstock(Q) + Cprod \qquad \textit{Equation 19}$$

The Csetup(Q), Cstock(Q) are plotted in Figure 15. To find the minimum point (Qopt), where the cost for the set-up and cost for the stock is optimised, Equation 19 is derived with respect to the quantity Q:

$$\frac{dCtotal(Q)}{dQ} = -\frac{D * Csetup}{Q^2} + \frac{Cstock}{2} \qquad \textit{Equation 20}$$

To find the minimum point, set Equation 20 to zero and rearrange the equation as

$$Qopt = \sqrt{\frac{2 * D * Csetup}{Cstock}} \qquad \textit{Equation 21}$$

The optimal point is presented in Figure 15.

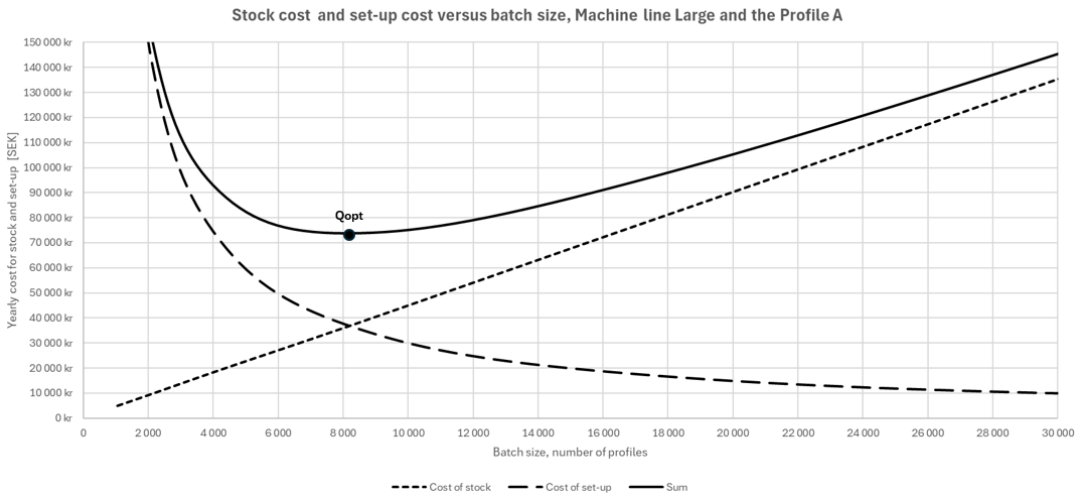


Figure 15. Cost for stock and set-up and the optimal batch size. In this case about 8200 profiles.

6. Results

6.1. Roll forming process

In this section, the relationship between average cost and batch size is presented for all profile types across the different machine line configurations. The analysis is structured in three steps. First, the various profile types are compared when produced on the same machine line; Large, Medium, and Small, to illustrate how economies of scale differ by profile complexity and production setup. Second, the focus shifts to comparing how a single profile type performs across different machine line options, highlighting how machine capability and investment level influence cost efficiency. Lastly, the cost of storing finished profiles is incorporated into the analysis, providing a more comprehensive view of the total cost structure and its sensitivity to production volume.

6.1.1. Average cost of producing with roll forming, without costs for storage

The graphs, Figure 16 to Figure 18, illustrate the relationship between batch size and average unit cost for several different profile types produced with the roll forming option: Large, Medium and Small, respectively. A clear pattern emerges: all curves show a sharply falling average cost at low batch sizes, followed by a gradually decreasing cost reduction as the batch size increases. This is a classic expression of economies of scale, where fixed costs can be distributed over a larger number of units produced. As the batch size continues to increase, the curves flatten out. This indicates that variable costs – such as materials, energy and ongoing operations – dominate the cost picture at large volumes. Since these costs are proportional to the number of units produced, the average cost decreases only marginally as the batch size increases further. This decreasing margin power is typical for processes with high fixed costs and relatively low variable costs, which is characteristic of roll forming.

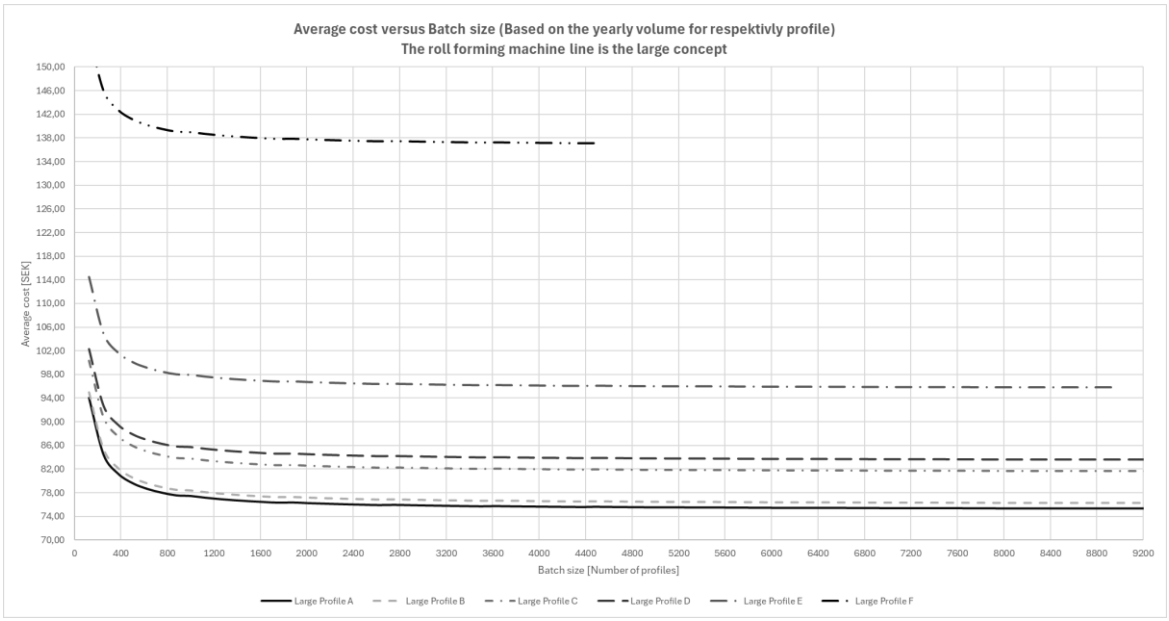


Figure 16. Machine line concept Large and the average cost for different profiles vs. batch size.

Finally, the analysis shows that the cost advantages of roll forming are strongly dependent on the batch sizes – and thus the annual volumes – being high enough for the set-up cost to be distributed over many units produced. For the three profiles with the lowest annual volumes, this means that the average cost remains high even with increasing batch sizes, as the fixed set-up cost and tooling investment cannot be sufficiently eliminated. In these cases, the economies of scale that make roll forming competitive are

missing. Therefore, bending is a more suitable manufacturing option for these profiles, as the process has significantly lower set-up costs and does not require the same volume to achieve cost-effectiveness. This is particularly apparent for the machine concept Large in Figure 16, where the cost of the machine and tooling is so high that it requires high volume.

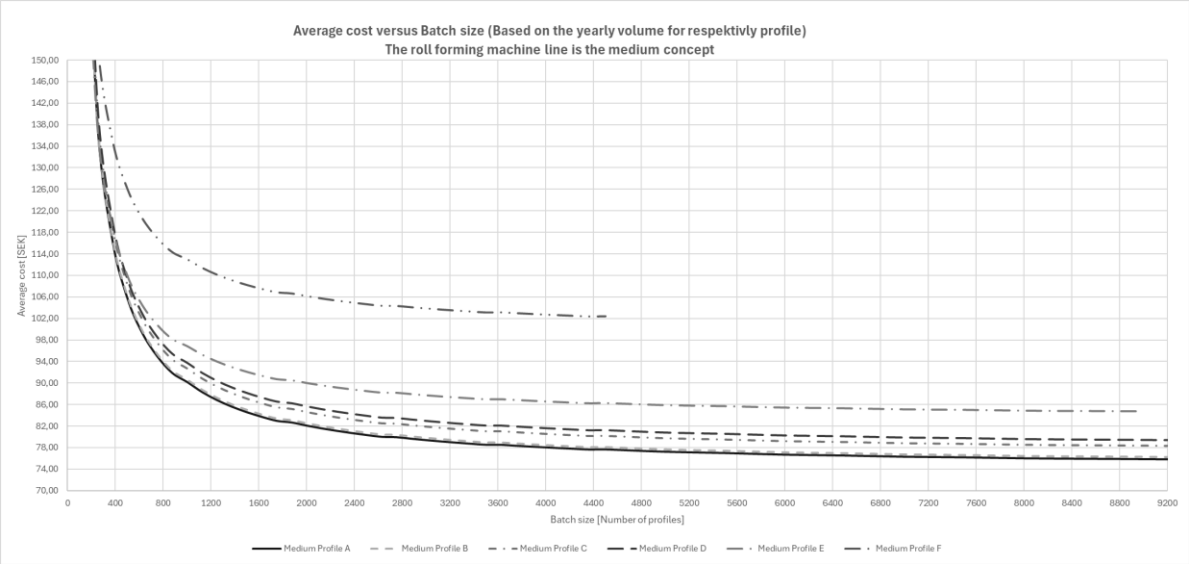


Figure 17. Machine line concept Medium and the average cost for different profiles vs. batch size.

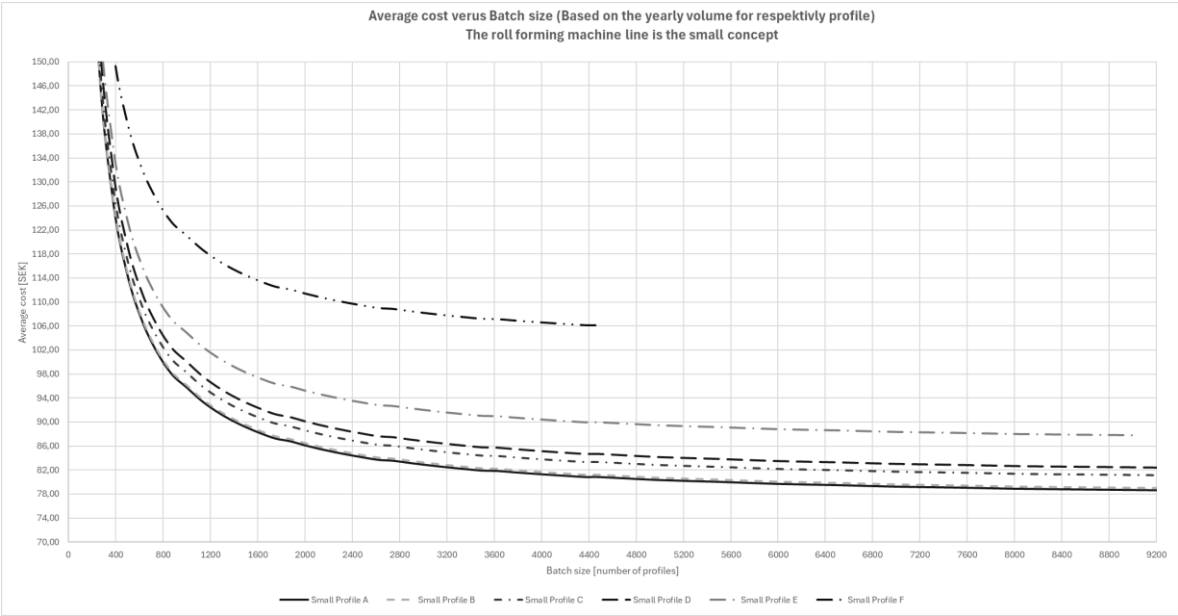


Figure 18. Machine line concept Small and the average cost for different profiles vs. batch size.

6.1.2. Roll forming process, comparison between the concepts

The graphs show, Figure 19 and Figure 24 (Appendix A), how the average unit cost of the “Profile A” and “B” develops at different batch sizes when the profile is produced on three different roll forming lines: Small, Medium and Large. Since this profile has the highest annual volume in the study, it is particularly relevant for analysing how the machine line's capacity and set-up cost affect cost efficiency.

A clear pattern emerges: the large line is the most cost-effective for small and medium-sized batch sizes. This is because the large line has shorter set-up times and thus lower set-up costs per batch. In the case of small batches, where the set-up cost makes up a large proportion of the total cost, this has a significant effect on the average cost. The faster changeover means that the cost curve for the large line is lowest at the beginning of the range and falls faster than the other alternatives.

However, as the batch size increases, the cost relationship between the machine lines changes. For large batch sizes, the medium line will be the cheapest option, despite the fact that it has a higher set-up cost than the large line. The explanation is that the medium line has a lower investment cost and thus a lower cost of capital per unit produced. When the batches become large enough for the set-up cost to have a marginal significance, the machine line's fixed cost dominates, which makes the medium line more cost-effective in the upper volume range.

The Small line is consistently the most expensive option, both for small and large batch sizes. Despite being the cheapest to buy, it has two structural disadvantages: high set-up cost and lower production speed. The combination of long changeover time and lower capacity means that the average cost of the small line never competes with the medium or large line, regardless of batch size. This clearly illustrates that a low investment cost is not enough to achieve cost-effectiveness in high-volume production.

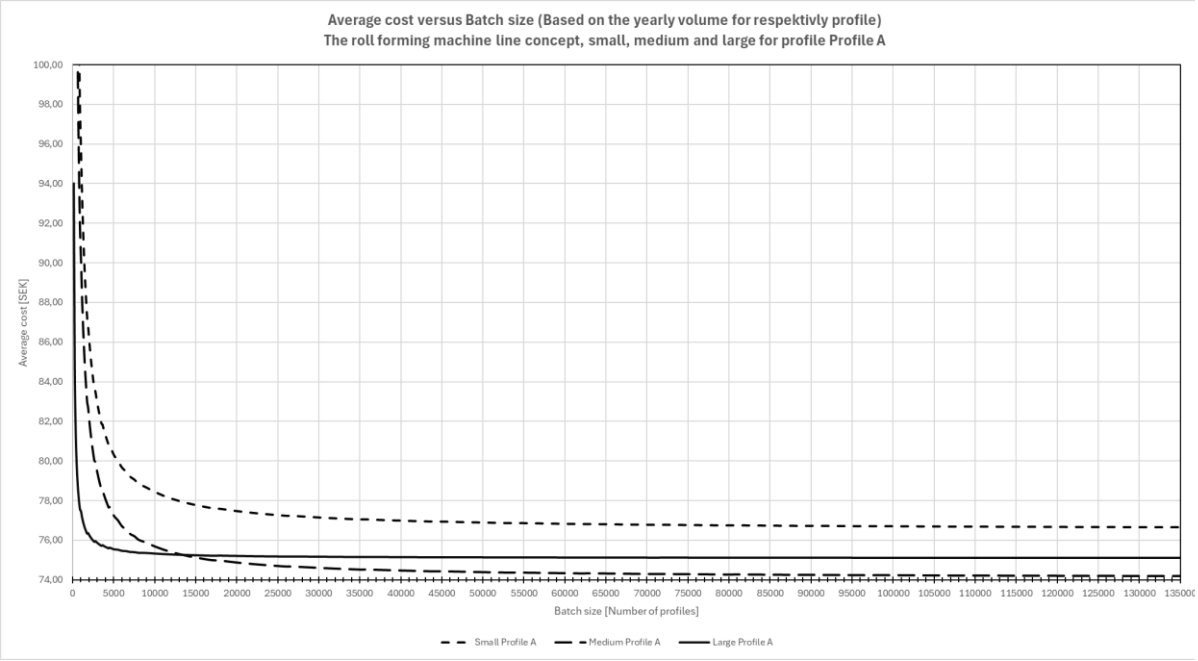


Figure 19. "Profile A" produced with the different machine concept.

In an alternative where the annual volume is only 36,000 profiles, Figure 25 (Appendix A), the cost picture changes significantly. At such low volumes, the large line quickly becomes an expensive choice. Despite its high capacity, both the cost of capital and the cost of tools are too high in relation to the limited production. The fixed cost base is increasingly distributed among units, which means that the average unit cost rises sharply and never reaches a competitive level.

In this lower volume scenario, the small machine option will instead be more competitive than before. The lower investment cost means that the capital cost per unit produced is significantly lower than for the large line, and even if the set-up time is long and the production speed is lower, these factors have

less impact when batch sizes are small, and the total annual volume is limited. The Small line can therefore perform better relatively than in the high-volume cases.

Despite this, the medium line remains the most cost-effective option even with an annual volume of 36,000 profiles. It offers a balance between reasonable investment costs and good production capacity, which allows both capital cost and cost of tools to be maintained at a level that gives the lowest average unit cost in this volume range. The medium line thus becomes the alternative that best handles the combination of low volume and requirements for cost-effectiveness.

6.1.3. Roll forming process, the optimum batch size

When inventory cost is included in the costing model, the economic logic behind batch size selection changes significantly. In the previous analyses (Figure 19), it was shown how the average production cost drops rapidly at small batch sizes and then levels off when the fixed costs for machine, tooling and changeover are distributed over more units produced. This cost structure creates clear economies of scale, where larger batches result in lower average costs as long as the production cost is considered in isolation.

However, when the cost of inventory is added, the cost picture changes. The graph presented here, see Figure 20 and Figure 26 - Figure 27 (Appendix A), shows how the total average cost – production plus inventory – develops as the batch size increases. The result is that the previously constantly falling cost curve now takes on a U-shape, which is a classic result from the EOQ model, chapter 5.4.4. With the help of the model, the optimum batch size can be calculated. The result is presented for the different profiles combined with the different machine concepts in the Table 5. The optimal batch size and the Minimum Efficient Scale (MES) point are presented in the Figure 20 and Figure 26 - Figure 27 (Appendix A).

Table 5. The optimal batch size for different profile and machine alternative.

Line type and profile	Production cost	Yearly demand	Cost for storing	Stock cost per unit	Set-up cost per batch size	Optimal batch size
Large Profile A	75,10	135000	12%	9,01	2229	8172
Medium Profile A	74,20	135000	12%	8,90	14665	21088
Small Profile A	76,65	135000	12%	9,20	17920	22935
Large Profile B	76,05	90000	12%	9,13	2229	6631
Medium Profile B	74,65	90000	12%	8,96	14665	17166
Small Profile B	77,11	90000	12%	9,25	17920	18671
Large Profile C	77,30	36000	12%	9,28	2229	4159
Medium Profile C	77,03	36000	12%	9,24	14665	10688
Small Profile C	79,54	36000	12%	9,54	17920	11627

The figures show that as the batch size increases, the average production cost initially decreases, as in Figure 20. However, after a certain point, the total average cost starts to rise again. This is because the cost of inventory is growing faster than the cost of production is falling.

This means that for small batch sizes, the changeover cost dominates, which means that the average cost is high, for medium batch sizes, the downward trend of production costs is balanced by a moderate inventory cost. This is where the lowest total cost arises, the economically optimal batch size. With large batch sizes, the cost of storing profiles increases sharply, driving up the overall average cost even though the cost of production continues to fall marginally.

This is an important result, because it shows that the batch size that minimises the production cost is not the same as the batch size that minimises the total cost.

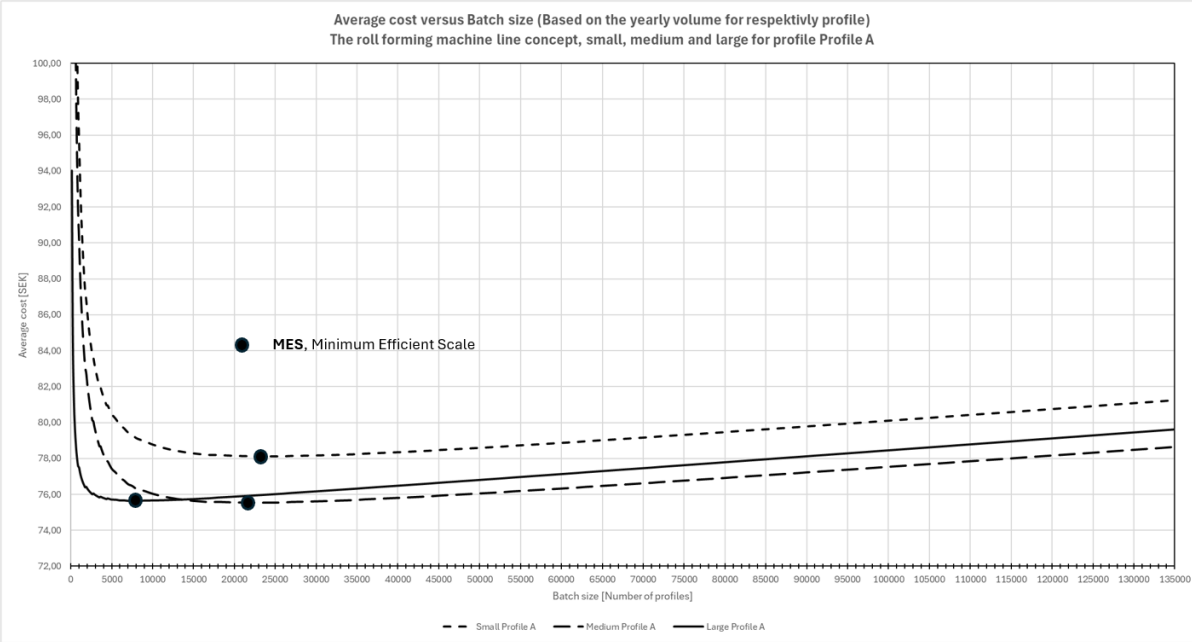


Figure 20. Average cost of "Profile A" for different machine concept with cost for storing included.

For "Profile C", see Figure 27 (Appendix A), which has a relatively low annual volume of 36,000 profiles; the cost structure shows the same basic pattern both with and without the cost of storing. Already in the analysis of the production cost alone, it was clear that the large roll forming line is not an economically rational choice for this profile. The tool cost and the cost of capital are simply too high in relation to the limited volume, which means that the large line never reaches a competitive average cost.

6.1.4. Roll forming production time at the optimal batch sizes

Table 6 shows clear differences in both optimal batch size and total production time across the three machine concepts. The large line achieves the lowest accumulated production time due to short changeover times and high production speed, making it highly efficient when volumes are sufficiently high. In contrast, the medium line can produce at the same speed as the large line, which results in similar total production times. However, its higher changeover cost leads to a higher total cost per batch. The small line performs the weakest: it has both significantly higher changeover costs and a slower production rate, which together result in substantially longer production times and higher total costs compared with the large alternative.

These cost and time differences also influence how much unused capacity each machine concept has. With approximately 1,750 available production hours per year on a single shift, both the large and medium lines have considerable room to absorb increased volumes without becoming bottlenecks. The Small line, however, has limited capacity for additional production due to its long cycle times and higher setup requirements. Overall, the results indicate that the large line becomes increasingly advantageous as volumes grow, while the medium line remains a balanced option with competitive production speed but higher setup costs. The Small line is only suitable for low-volume scenarios where its slower pace and higher setup cost do not restrict capacity or drive costs excessively.

Table 6. Time for production of different profile type and machine concept.

Machine concept	Profile type	Optimal bacth size	Number of batches	Production time per batch (Hour)	Total production time [Hour]	Ackumulated time [Hour]	Total cost, storing and production	Ackumulated cost [SEK]
Large	Profile A	8 172	17	11	181	181	73 646	
Large	Profile B	6 631	14	9	123	305	60 511	
Large	Profile C	4 159	9	6	52	**357**	38 584	**172 740**
Medium	Profile A	21 088	6	35	226	226	187 765	
Medium	Profile B	17 166	5	30	156	382	153 774	
Medium	Profile C	10 688	3	20	68	**450**	98 793	**440 333**
Small	Profile A	22 935	6	98	579	579	210 959	
Small	Profile B	18 671	5	82	396	976	172 763	
Small	Profile C	11 627	3	56	172	**1148**	110 973	**494 695**

6.2. Profile bending process

In this section, the economic performance of the PB process is analysed across the three levels of automation: Small (manual), Medium (semi-automated cell), and Large (fully integrated line). Unlike the RF process, which relies on high-volume batches to amortise setup costs, the PB process is characterised by its high flexibility and near-zero setup times (for Medium and Large concepts) in automated configurations. Therefore, the analysis shifts from batch size optimisation to the realisation of economies of scope and the ability to fulfil on-demand (JIT) production.

6.2.1. Average cost for producing with the profile bending process

The average cost is simply calculated using Equation 8 and includes fixed and variable costs per unit, including labour, materials, energy, and SC (Semi-Variable Costs/Additional Costs), additional fixed costs or a specific category of overhead. The result is summarised and presented in Figure 21.

The analysis of the three concepts showed that the AC for profile bending was coupled to the profile complexity rather than quantity. In the Small (manual) case, the process is dominated by high variable labour costs due to the intensive manual handling required for each bend. In the Large (automated) case, high initial investment results in significant fixed costs (FC), but these are offset by the lowest variable costs (VC) as the system can operate unattended with minimal waste.

A critical finding is that for complex profiles (e.g., “Profile C”), the variable cost in profile bending increases because the cycle time is a direct function of the number of bends. However, because profile bending uses universal tooling, it avoids the high product-specific tool investments required for roll forming, maintaining a more stable cost structure across a diverse product mix.

6.2.2. Profile bending process, comparison between the concepts

Comparing the three levels of profile bending equipment reveals a strategic trade-off between capital intensity and operational flexibility. A comparison of the calculated profile cost is shown in Figure 21.

Large Concept: This fully integrated system achieves the lowest AC at high total annual volumes (approaching the 500,000-meter scenario). Its automatic tool changers allow for a seamless transition between the seven different profile types in the product mix without manual intervention.

Medium Concept: This semi-automated cell provides a balanced position, offering higher precision and faster changeovers than manual methods, but with lower capital risk than the large concept. It is the most cost-effective solution for moderate annual volumes where the high fixed cost of the large line cannot be fully amortised.

Small Concept: The manual press brake has the lowest fixed cost base, making it competitive only at very low volumes or for prototyping. Its high reliance on operator expertise and slow production speeds make it unsuitable for the total volume requirements of 100,000 frame sets.

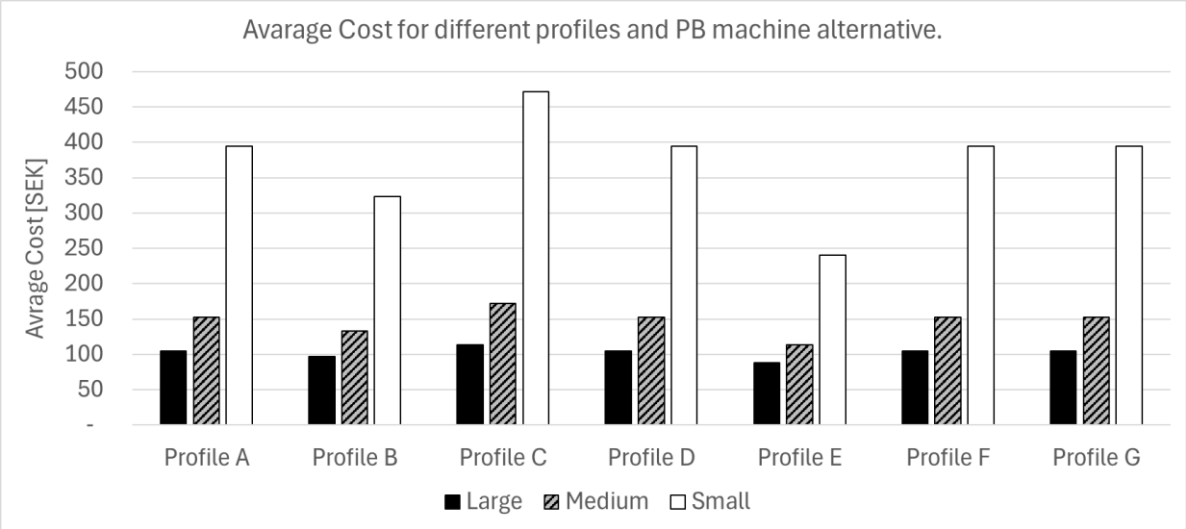


Figure 21. The average cost for different profiles and profile bending machine alternatives

In Figure 21 there is a large difference in the resulting profile costs between PB Small and Medium concepts, indicating the importance of at least entry-step automation level and the benefit of not needing to flip the profile for reverse bends. In contrast the gap between PB Medium and Large concepts is considerably smaller in comparison to the respective difference in machine cost, suggesting that there is a limit in this technology, where additional investments yield decreasing returns in terms of efficiency.

6.2.3. Profile bending, average cost in relation to complexity

The production time for profile bending is calculated based on the cycle time per part, which varies with profile complexity, see Figure 22. As shown in a bend program simulation of profiles, the total production time for the large concept is the lowest due to integrated material handling and high-speed folding beams.

Relying purely on traditional, static cost accounting models often obscures hidden operational expenses generated during complex bending cycles (Janovska et al. 2022). Thus, implementing life cycle costing can drive life cycle management forward by anchoring a long-term economic mindset within engineering design decisions. Methodological design choices in LCC are often shaped by a continuous interplay between theoretical ideals and alignment with established organizational routines. An over-reliance on purely financial indicators within life cycle costing can sometimes narrow a firm's perspective, overshadowing broader long-term sustainability metrics (Kambanou 2020).

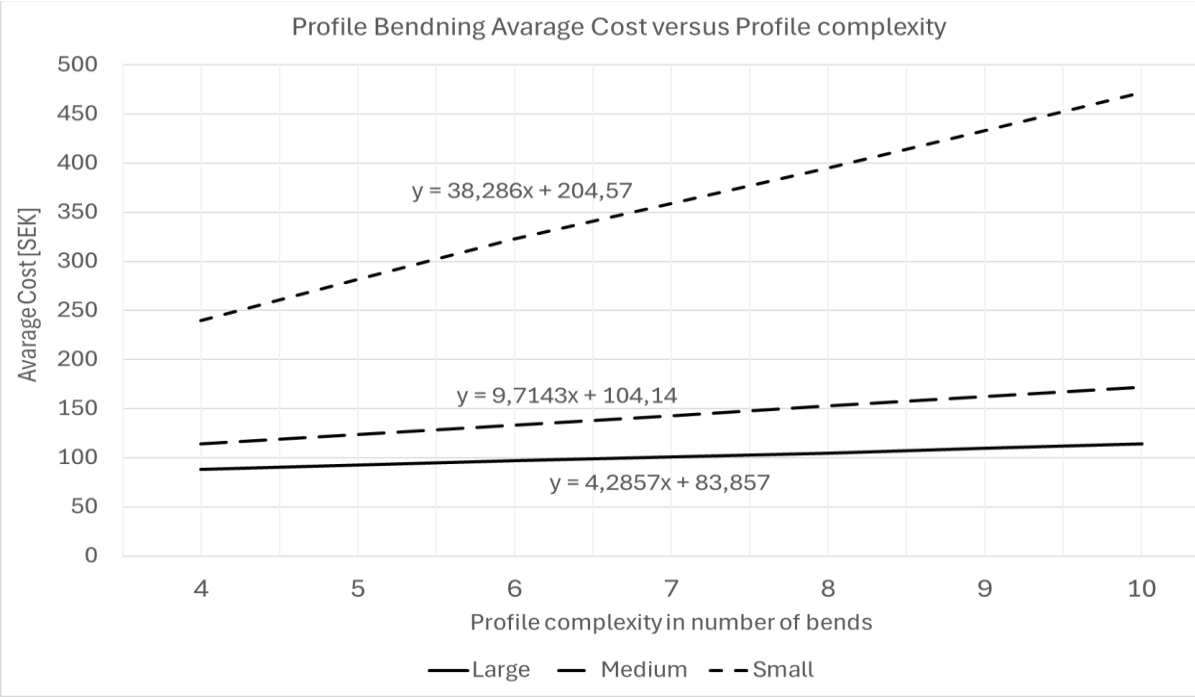


Figure 22. The average cost in relation to profile complexity (number of bends).

6.2.4. Profile bending production time and capacity utilisation

The production time for profile bending is calculated based on the cycle time per part, which varies significantly with profile complexity. As shown in the simulation for 500,000 meters of profile, the total production time for the large concept is the lowest due to integrated material handling and high-speed folding beams.

While the RF process is superior in absolute production speed (meters per minute), the PB process offers higher Overall Equipment Effectiveness (OEE) in environments with high variety. The absence of long tool changeovers (which can take days in RF without cassette systems) means that the PB Large line has a much higher available production time for actual manufacturing, making it the superior choice for the "high-variety, low-to-medium volume" segments of the frame market.

7. Discussion

7.1. The roll forming process

The results from Chapter 6.1 clearly show that the roll forming process is dominated by economies of scale, which is fully consistent with the theoretical framework presented in the literature review. Across all profiles and machine concepts, the relationship between average cost and batch size follows an L-shaped pattern, where costs decline rapidly at low batch sizes and gradually flatten as volume increases. This cost behaviour corresponds closely to the long-run average cost (LRAC) curves discussed by Stigler (1958), Silberston (1972), and Besanko et al. (2017), and confirms that roll forming is a scale-intensive production process.

The analysis demonstrates that the high capital intensity of roll forming is not a weakness but a key driver of cost efficiency when volumes are sufficiently large. As shown in se Figure 16 - Figure 18, fixed costs related to machinery, tooling, and automation are distributed across a growing output as batch size increases, leading to sharp reductions in average cost until a practical Minimum Efficient Scale (MES) is reached. This aligns with Silberston's (1972) observation that most manufacturing industries exhibit strong cost reductions well before MES, followed by a wide cost plateau. The results therefore, reinforce the relevance of MES as an operational benchmark rather than a precise cost-minimising point.

The comparison between the Small, Medium, and Large machine concepts further illustrates Stigler's (1958) argument that scale advantages are primarily determined by technology and plant design rather than investment size alone. The small roll forming line, despite its low investment cost, consistently suffers from high set-up costs and low production speed, resulting in structurally high average costs. This supports the view that low capital investment does not guarantee competitive production if capacity utilisation and set-up efficiency are insufficient. In contrast, the large line achieves strong economies of scale through short changeover times and high automation, but only under conditions of high annual volume, illustrating how scale advantages can turn into cost disadvantages when demand is misaligned with installed capacity.

Across most scenarios, the medium machine concept emerges as the most cost-efficient alternative. Its combination of moderate investment cost, relatively high production speed, and reduced set-up time allows it to operate close to the practically relevant MES across a wide range of batch sizes. From an industrial economics perspective, this outcome supports Stigler's survivor principle, which suggests that multiple firm or technology sizes can coexist efficiently, depending on how well they match market demand and production conditions.

When inventory costs are incorporated, the cost structure changes in a way that further strengthens the link between theory and results. The U-shaped total cost curves shown in Figure 20 and Figure 26 - Figure 27 (Appendix A) demonstrate that the batch size minimising production cost differs from the batch size minimising total cost. This confirms the argument by Besanko et al., (2017) that inventory effects and tied-up capital must be considered when evaluating economies of scale. Particularly for profiles with lower annual volumes, inventory costs quickly outweigh marginal production cost savings, which limits the economically optimal exploitation of scale.

From an economy of scope perspective, the analysis confirms the theoretical expectation that roll forming remains a relatively inflexible process. Each additional profile requires dedicated tooling and set-up efforts, which constrains cost-effective product variety, as described by Panzar and Willig (1981). While technical solutions such as cassette systems significantly reduce set-up costs and improve flexibility, they do not alter the fundamentally scale-oriented nature of the process.

Overall, the discussion confirms that roll forming is a production process where economies of scale dominate cost performance. The results from Chapter 6.1 strongly support the theoretical insights presented in Chapter 3, while also showing that the realisation of these scale advantages is highly sensitive to batch size, annual volume, and investment level.

7.2. The profile bending process

The findings from the PB process analysis confirm the fundamental shift in economic logic compared to RF. Whereas roll forming relies on distributing product-specific fixed costs across high volumes, the profile bending process demonstrates economies of scope using quasi-public inputs (Panzar & Willig, 1981). The following discussion critically examines these findings within the frameworks of mass customisation, subadditivity, and strategic positioning.

The primary advantage of the PB process lies in its inherent subadditivity of costs. Results indicate that the total cost (TC) of producing seven different profile types jointly on a single automated PB line is lower than the combined cost of producing them on separate, specialised lines (Panzar & Willig, 1981).

This synergy results from the machine's universal tooling, which serves as a shared input not fully consumed by any single product line, according to Besanko et al. (2017). This feature enables firms to avoid the tooling penalty that constrains roll forming as product variety increases. However, this advantage depends on Porter's (1987) "Better-Off Test", which requires that shared activities generate a tangible competitive advantage rather than superficial synergies. Additionally, Besanko et al. (2017) note that managing seven different profiles on one machine can introduce administrative complexity, potentially offsetting technical efficiency gains with agency inefficiencies.

Within the study's frame scenario, the PB process serves as the primary enabler of mass customisation, delivering individually tailored products at unit costs comparable to mass production (Pine et al., 1993). By eliminating tool changeovers, PB facilitates customisation at scale, making lot sizes of one economically viable (Blecker & Abdelkafi, 2006). This is achieved through delayed product differentiation, in which sheet metal remains generic until a customer order initiates the specific bending sequence at the Customer Order Decoupling Point (CODP) (Blecker & Abdelkafi, 2006).

The findings indicate that the PB Large (automated) concept achieves a flatter average cost (AC) curve than roll forming. This outcome implies that PB is less affected by batch size volatility, making it well-suited for Just-in-Time (JIT) strategies that reduce the cost of capital tied up in inventory (Pine et al., 1993). In contrast to roll forming, which relies on an Economic Order Quantity (EOQ) model to manage high setup costs, PB enhances the value chain by mitigating inventory risk. Nevertheless, a significant limitation persists: the PB process operates in a step-by-step manner. As profile complexity increases, variable costs rise linearly, resulting in a complexity penalty that is not present in roll forming, where production speed remains relatively constant regardless of profile intricacy.

Selecting among the PB Small, Medium, and Large concepts exemplify the managerial trade-off between capital intensity and operational flexibility (Keat et al., 2014). The PB Small (manual) scenario represents a low-fixed-cost, high-variable-cost technology that is competitive only at very low volumes, where capital requirements are the main constraint. Although the PB Large concept involves high capital investments, it is set up to achieve benefit in terms of both cost, lead time and flexibility leadership by providing a wide variety of unit costs comparable to mass production according to Besanko et al. (2017).

Strategically, the PB Large line enables a firm to avoid the risk of being stuck in the middle by pursuing a focused strategy targeting high-variance, customised market segments Besanko et al. (2017). Through flexible automation, the firm can achieve high Overall Equipment Effectiveness (OEE), as the machine spends more time producing and less time in idle changeover states.

In summary, although roll forming remains the leading technology for achieving Minimum Efficient Scale (MES) in high-volume commodity production (Besanko et al., 2017), the profile bending process provides a robust alternative for markets requiring strategic flexibility. Integrating PB technology enables firms to approach the efficiency frontier by optimising cost structures for diversity rather than solely for volume. However, industrial managers should consider the capacity limitations of PB, as its absolute throughput per shift remains lower than that of a high-speed RF line. Consequently, the optimal economic choice depends on Strategic Fit, as outlined in Besanko et al. (2017), requiring alignment between the selected technology, market demand, and internal capabilities.

7.3. Mix of the process and further considerations

The comparison between roll forming and profile bending in a mixed production context highlights the fundamental difference between a scale-efficient and a scope-oriented manufacturing process. Roll forming exhibits a strongly volume-dependent cost structure, where the average cost per unit decreases rapidly as batch size increases, whereas profile bending is characterised by a more stable, but generally higher, average cost at larger volumes. This reflects that roll forming is dominated by economies of scale through high fixed costs and low variable costs, while profile bending primarily relies on economies of scope due to low setup costs and high flexibility. At low batch sizes, profile bending is therefore the more cost-effective alternative; however, as batch size increases, distinct break-even points emerge where roll forming becomes the economically superior process.

This relationship is clearly illustrated in Figure 23, where several break-even points can be identified depending on batch size and machine concept. Beyond these points, the cost advantage shifts in favour of roll forming, indicating that a minimum production volume is required to justify the higher fixed and setup costs associated with the roll forming process. From a production planning perspective, this implies that roll forming should only be selected when batch sizes exceed the relevant break-even level. However, operating roll forming at batch sizes as small as approximately 1,000 profiles is primarily economically viable when using the Medium or Large machine concept, where setup and changeover costs between different profiles can be significantly reduced through shorter set-up times. Under these conditions, producing consolidated batches corresponding to full coils—approximately 1,000 profiles—allows the firm to exploit economies of scale while maintaining acceptable flexibility within a mixed production environment.

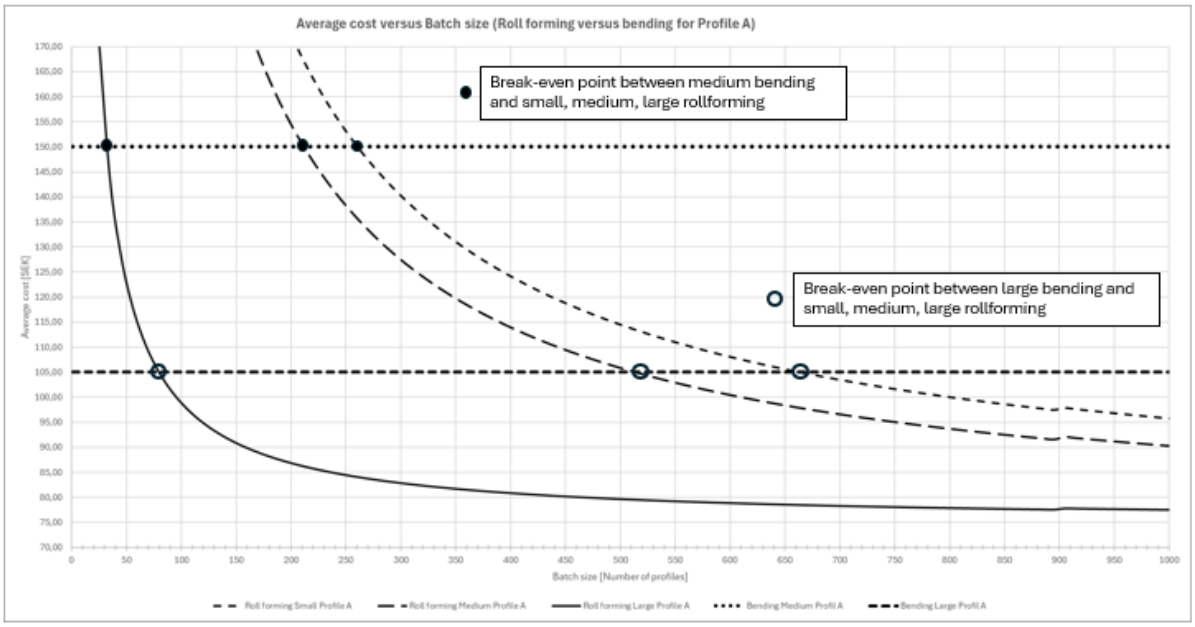


Figure 23. RF vs. PB for Profile A. Break-even point for different batch size and machine concepts.

From a growth and capacity-planning perspective, the results indicate that a mixed production strategy offers clear strategic advantages over relying solely on profile bending. In the current situation, where all profiles are produced using bending, high-volume products such as Profile A and Profile B consume a disproportionate share of available production time. As total demand increases, this creates capacity pressure and limits the firm's ability to respond flexibly to new products or low-volume variations. The analysis suggests that, beyond the identified break-even points, investing in a roll forming line for these high-volume profiles becomes economically justified, particularly when production can be shifted to larger, consolidated batch sizes. By allocating Profile A and Profile B to roll forming, the firm can exploit economies of scale while stabilising unit costs as volume grows.

At the same time, introducing roll forming does not replace the need for profile bending. but rather complements it. By offloading high-volume, standardised profiles to a roll forming line (preferably using a Medium or Large machine concept) with reduced setup and changeover costs, substantial capacity is freed up in the existing bending equipment. This released capacity can then be used to focus on low-volume, high-variety products, where profile bending's economies of scope and flexibility provide the greatest economic value. In this way, the firm can simultaneously pursue volume growth and maintain strategic flexibility, avoiding excessive investment in scale across the entire product portfolio. The proposed mix of processes therefore supports a scalable growth strategy, where roll forming enables cost-efficient expansion in selected product segments, while profile bending remains a key enabler for customisation, product development, and responsiveness to changing market demand.

8. Conclusion and Future Research

This thesis has analysed the economic trade-offs between roll forming and profile bending by integrating theories of economies of scale and scope with a quantitative life-cycle cost analysis. The results demonstrate that the two processes exhibit fundamentally different cost structures and strategic roles within sheet metal profile manufacturing. Roll forming is clearly a scale-intensive process, where high fixed and setup costs can be efficiently distributed only at sufficiently large batch sizes and annual volumes. Profile bending, in contrast, is dominated by economies of scope and provides substantial flexibility through low setup costs, universal tooling, and the ability to handle product variety and short production runs.

A central conclusion of the study is that neither process should be viewed as a universally superior solution. Instead, the economically optimal choice depends on production volume, batch-size strategy, and product mix. For low-volume and high-variety profiles, profile bending remains the most cost-effective alternative. However, for high-volume profiles such as Profile A and Profile B, the analysis shows distinct break-even points beyond which roll forming becomes the economically preferable process. In a growth scenario, continuing to produce these high-volume profiles using profile bending leads to capacity constraints and limits the firm's ability to respond flexibly to new or low-volume products. The results therefore support a mixed production strategy, in which investments in a roll forming manufacturing line (preferably a Medium or Large machine concept) are used to absorb volume growth. This approach exploits economies of scale for standardised, high-volume profiles while simultaneously freeing capacity in existing bending equipment to enhance flexibility and support economies of scope.

Table 7 summarises the main decision logic derived from the analysis, illustrating how production volume, product mix variation, and investment level jointly determine the economically preferred manufacturing process.

Table 7. Different investment depending on volume and product mix.

Production volume	Product mix variation	Investment level	Economically preferred process	Strategic interpretation
Low	High	Low	Profile bending (Small)	Flexibility dominates; low fixed cost and scope advantages outweigh scale effects.
Low	High	Medium	Profile bending (Medium)	Improved productivity while maintaining scope; suitable for JIT and customization.
Medium	Medium	Medium	Profile bending (Medium) / (Large)	Transition region; break-even points depend on batch size and profile complexity.
Medium	Low	Medium	Roll forming (Medium)	Scale effects start to dominate; moderate investment enables cost-efficient batches.
High	Low	High	Roll forming (Large)	Strong economies of scale; high fixed cost justified by volume and stable demand.
High	High	High	Mixed strategy (RF + PB)	Standard profiles allocated to roll forming, variants handled by bending to preserve flexibility.

The study thereby contributes a structured decision-support framework for selecting and combining production processes based on economic performance rather than purely technical considerations. From a managerial perspective, the findings highlight how a deliberate mix of scale-oriented and scope-oriented processes can support both cost-efficient growth and strategic flexibility.

Although the analysis in this study is based on a specific case within the manufacturing of frame profiles, the developed cost models are applicable to other profile-producing industries with similar production logic, such as building components, logistics systems, and transport-related profiles. The analytical framework can be used to identify break-even points between scale-oriented and scope-oriented production processes based on key variables, including production volume, product mix variation, profile complexity, and investment level. The results therefore indicate that a combined production strategy often provides a robust balance between cost efficiency and flexibility beyond the empirical case analysed in this study. In addition to production equipment, managing manufacturing flexibility requires analysing the direct interplay between a factory's complete input, throughput, and output characteristics (Höse et al., 2023).

It is important to consider that in the modern era characterised by frequent market shifts, human operators remain the most flexible asset due to their unmatched cognitive adaptation and problem-solving capacities. Achieving elevated physical adaptability in factory environments demands both economically optimised manufacturing technology and strategy, as well as e.g. machine modularity, quick-connect logic, and functional decoupling, and in the end, it is the organisational workforce that will conceptualise, implement and maintain production efficiency over time (ElMaraghy et al., 2021).

Another key factor is that a well-thought-out product strategy goes hand in hand with the production strategy. Product platforms are a core operational method used by progressive manufacturers to fulfil diverse market demands while shortening production lead times. Implementing platform strategies allows managers to strike a sustainable balance between maximising product variety and containing overall system customisation effort. Reconfiguring and customising standardised product platforms at various manufacturing assembly nodes allows factories to efficiently manage high-variety output (Galizia et al., 2020).

Further research could extend the scope of this study by analysing alternative market strategies and their implications for the optimal mix of roll forming and profile bending. Different strategic paths, such as volume-driven growth, increased product customisation, entry into niche markets, or a combination of standardised and differentiated product offerings, are likely to place distinct demands on the production system. Evaluating how these strategies influence capacity requirements, investment decisions, and the balance between economies of scale and economies of scope would provide valuable insights for long-term production planning. Additionally, scenario-based analyses could be used to assess how changes in demand structure over time affect the economically optimal allocation of profiles between roll forming and bending. Such an approach would strengthen the decision-support framework by explicitly linking manufacturing process selection to corporate growth strategy and market positioning.

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Appendix A:

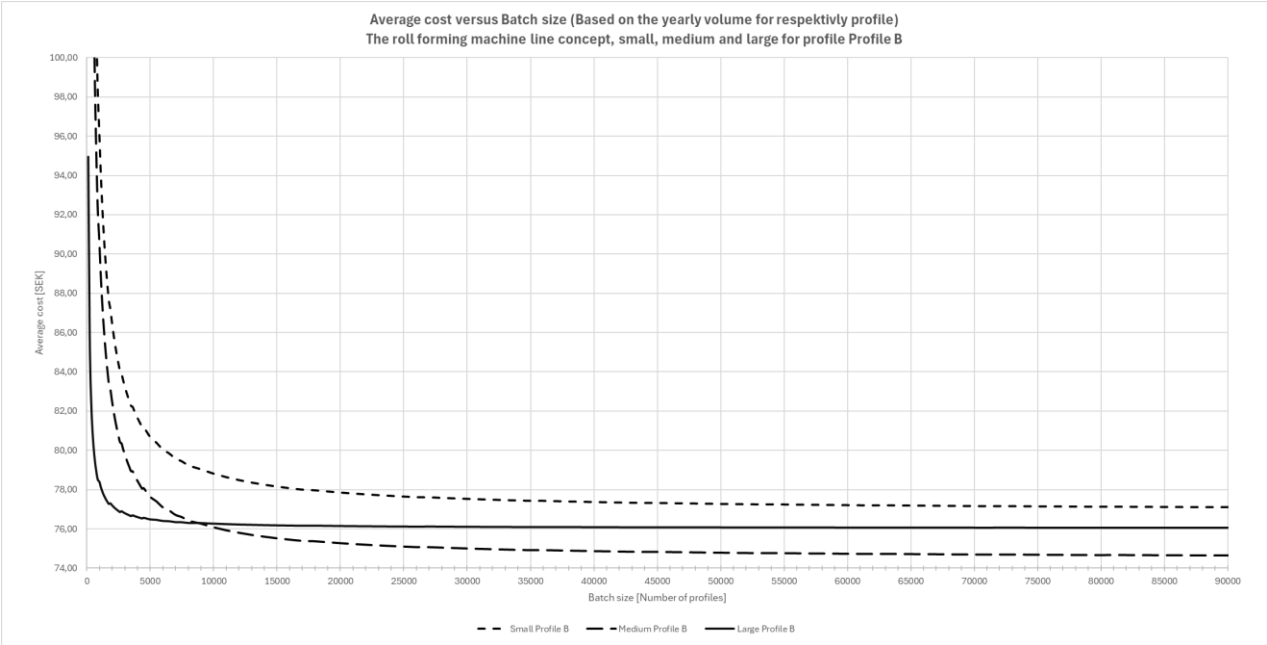


Figure 24. “Profile B” produced with the different machine concepts.

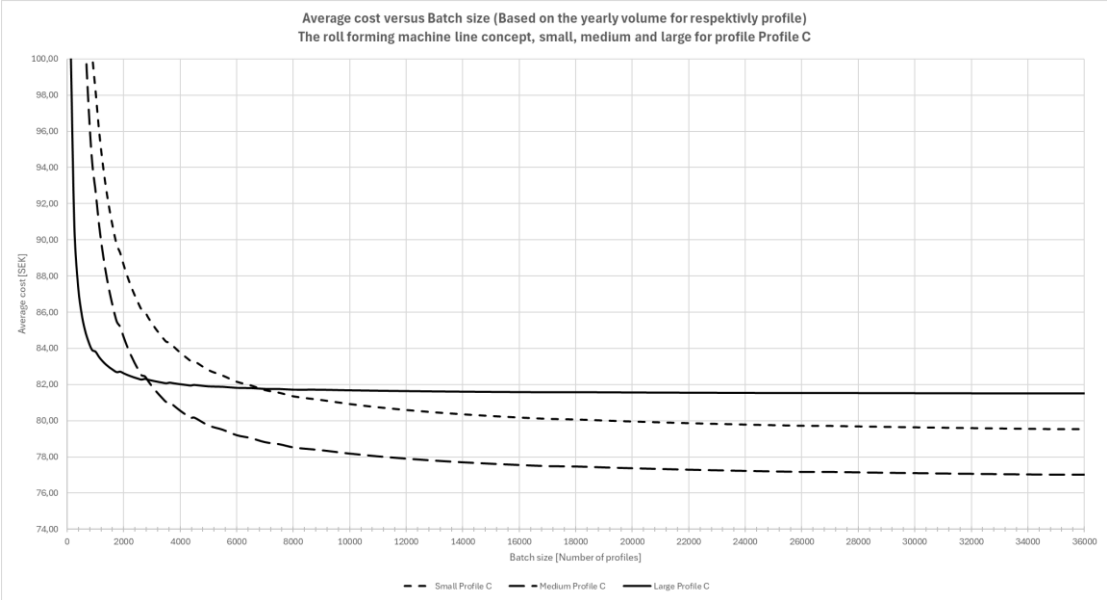


Figure 25. “Profile C” produced with the different machine concepts.

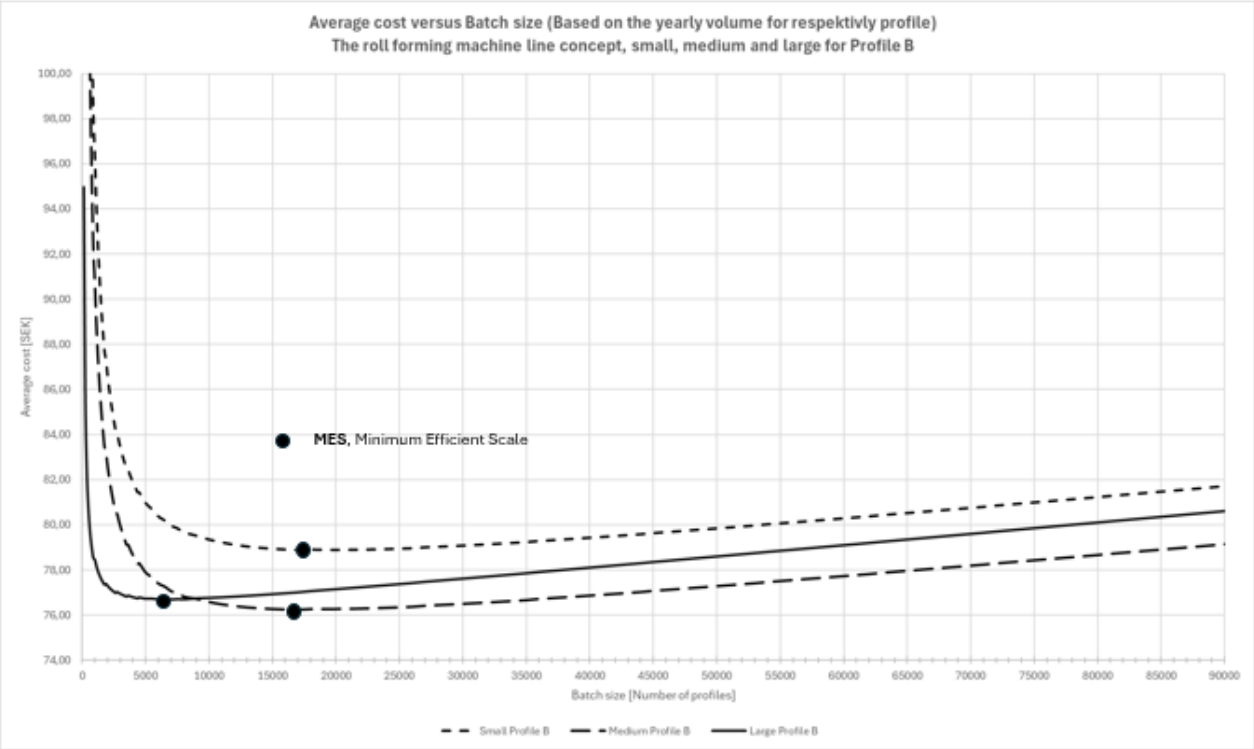


Figure 26. Average cost of "Profile B" for different machine concepts with storage cost included.

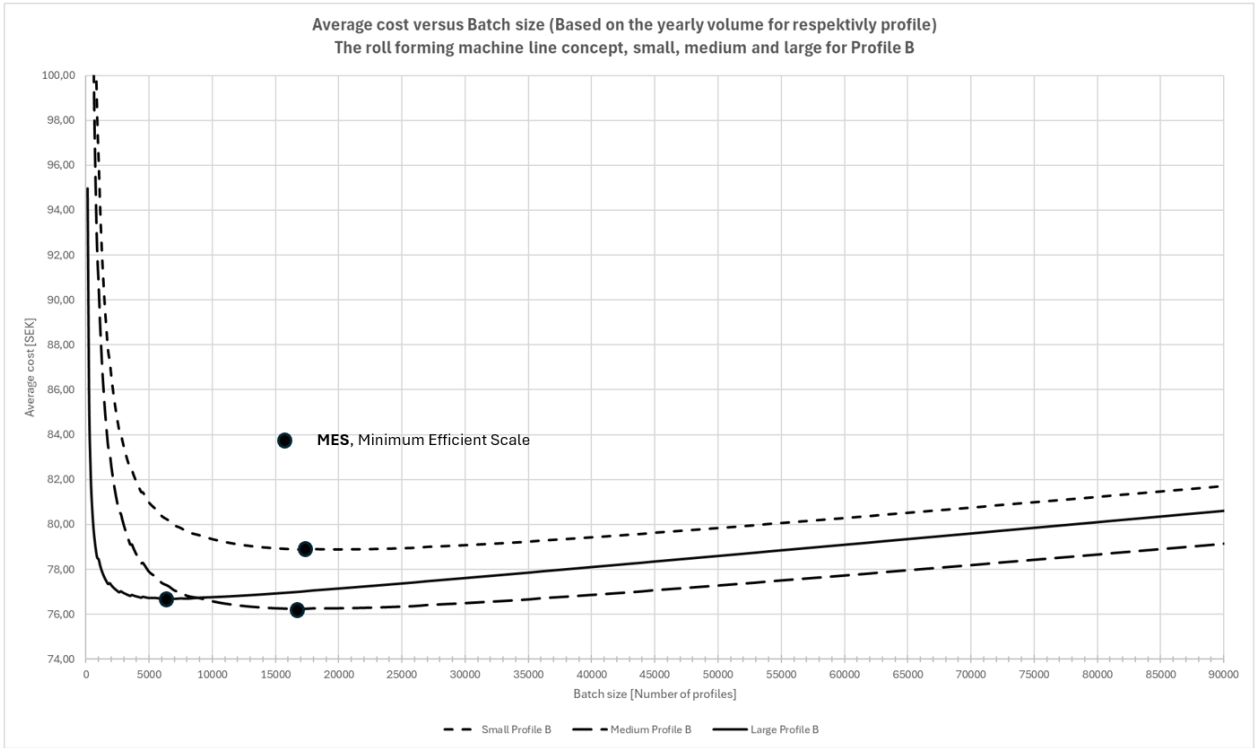


Figure 27. Average cost of "Profile C" for different machine concepts with storage cost included.

Appendix B:

Capital cost per hour for the respective investment: roll forming Small, Medium and Large, Table 8.

Table 8. Capital cost per roll forming machine concept.

	Small	Medium	Large
Investment	10 000 000	17 000 000	20 000 000
Depreciation time, linear	8	8	8
Resiudal value	0%	0%	0%
Capital cost, WACC	10%	10%	10%
Average tied-up capital	5 000 000	8 500 000	10 000 000
Depreciation per year	1 250 000	2 125 000	2 500 000
Capital cost per year	500 000	850 000	1 000 000
Total yearly capital cost	1 750 000	2 975 000	3 500 000
Production hours, yearly	1 750	1 750	1 750
Capital cost per hour	**1 000**	**1 700**	**2 000**

Capital cost hour for the respective investment: profile bending Small, Medium and Large, Table 9.

Table 9. Capital cost per profile bending machine concept.

	Small	Medium	Large
Investment	12 500 000	17 000 000	30 000 000
Depreciation time, linear	8	8	8
Resiudal value	0%	0%	0%
Capital cost, WACC	10%	10%	10%
Average tied-up capital	6 250 000	8 500 000	15 000 000
Depreciation per year	1 562 500	2 125 000	3 750 000
Capital cost per year	625 000	850 000	1 500 000
Total yearly capital cost	2 187 500	2 975 000	5 250 000
Production hours, yearly	1750	1750	1750
Capital cost per hour	**1250**	**1700**	**3000**